

MLS

Line guidance sensors

SICK
Sensor Intelligence.



Described product

MLS
MLSE-0200
MLSE-0300
MLSE-0400
MLSE-0500
MLSE-0600

Manufacturer

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Production location

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Legal information

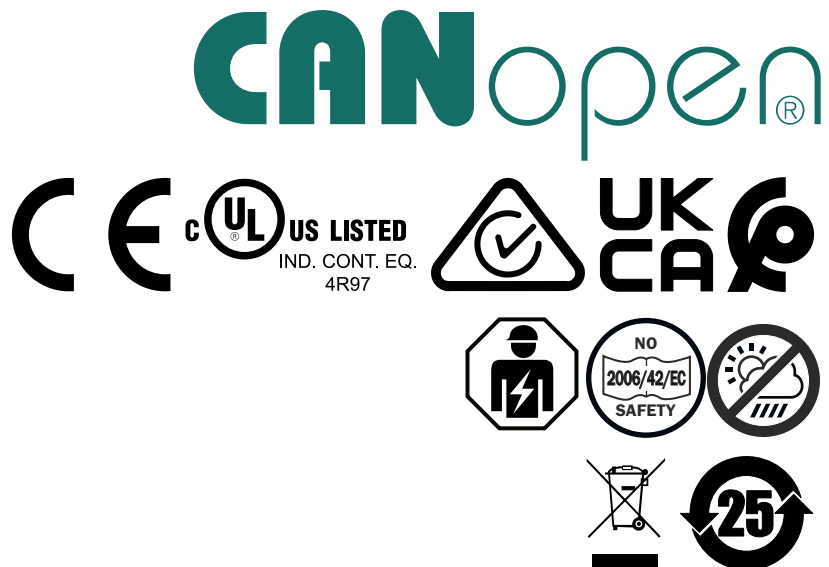
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Original document

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1 About this document

1.1 Information on the operating instructions

Read these operating instructions carefully before starting any work in order to familiarize yourself with the product and its functions.

The operating instructions are an integral part of the product and should remain accessible to the personnel at all times. When handing this product over to a third party, include these operating instructions.

These operating instructions do not provide information on the handling and safe operation of the machine or system in which the product is integrated. Information on this can be found in the operating instructions for the machine or system.

1.2 Scope

These operating instructions are used for incorporating a sensor into a customer system. Step-by-step instructions are given for all the actions required.

These instructions apply to all available device variants of the sensor.

Available device variants are listed on the online product page.

► www.sick.com/mls

Commissioning is described using one particular device variant as an example.

Simplified device designation in the document

In the following, the sensor is referred to in simplified form as “MLS”.

1.3 Symbols and document conventions

Warnings and other notes



DANGER

Indicates a situation presenting imminent danger, which will lead to death or serious injuries if not prevented.



WARNING

Indicates a situation presenting possible danger, which may lead to death or serious injuries if not prevented.



CAUTION

Indicates a situation presenting possible danger, which may lead to moderate or minor injuries if not prevented.



NOTICE

Indicates a situation presenting possible danger, which may lead to property damage if not prevented.



NOTE

Highlights useful tips and recommendations as well as information for efficient and trouble-free operation.

Instructions to action

- ▶ The arrow denotes instructions to action.
- 1. The sequence of instructions is numbered.
- 2. Follow the order in which the numbered instructions are given.
- ✓ The tick denotes the results of an action.

1.4 Further information

You can find the product page with further information via the SICK Product ID:
pid.sick.com/{P/N}/{S/N}
(see "Product identification via the SICK product ID", page 9).

The following information is available depending on the product:

- This document in all available language versions
- Data sheets
- Other publications
- CAD files and dimensional drawings
- Certificates (e.g., declaration of conformity)
- Software
- Accessories

1.5 Customer service

If you require any technical information, our customer service department will be happy to help. To find your representative, see the final page of this document.



NOTE

Before calling, make a note of all type label data such as type code etc. to ensure faster processing.

2 Safety information

2.1 Intended use

The MLS sensor is a non-contact sensor used to determine the position of a magnetic line tape.

SICK AG assumes no liability for losses or damage arising from the use of the product, either directly or indirectly. This applies in particular to use of the product that does not conform to its intended purpose and is not described in this documentation.



NOTICE

Radio interference may occur when the sensor is used in residential areas.

- ▶ Only use the device in industrial environments (EN 61000-6-4).

2.2 Improper use

- The sensor does not constitute a safety-relevant device according to the EC Machinery Directive (2006/42/EC).
- The sensor must not be used in explosion-hazardous areas.
- Any other use that is not described as intended use is prohibited.
- Any use of accessories not specifically approved by SICK AG is at your own risk.
- The sensor is not suitable for outdoor applications.



NOTICE

Danger due to improper use!

Any improper use can result in dangerous situations.

Therefore, take note of the following information:

- ▶ The sensor should be used only in line with intended use specifications.
- ▶ All information in these operating instructions must be strictly complied with.

2.3 Limitation of liability

Applicable standards and regulations, the latest state of technological development, and our many years of knowledge and experience have all been taken into account when assembling the data and information contained in these operating instructions. The manufacturer accepts no liability for damage caused by:

- Failing to observe the operating instructions
- Improper use
- Use by untrained personnel
- Unauthorized conversions
- Technical modifications
- Use of unauthorized spare parts, consumables, and accessories

With special variants, where optional extras have been ordered, or owing to the latest technical changes, the actual scope of delivery may vary from the features and illustrations shown here.

2.4 Qualification of personnel

Any work on the product may only be carried out by personnel qualified and authorized to do so.

Qualified personnel are able to perform tasks assigned to them and can independently recognize and avoid any potential hazards. This requires, for example:

- technical training
- experience
- knowledge of the applicable regulations and standards

2.5 Hazard warnings and operational safety

Please observe the safety notes and the warnings listed here and in other chapters of these operating instructions to reduce the possibility of risks to health and avoid dangerous situations.

2.6 Repairs

Repair work on the sensor may only be performed by qualified and authorized personnel from SICK AG. Interruptions or modifications to the sensor on the part of the customer will invalidate any warranty claims against SICK AG.

3 Product description

3.1 Product identification

3.1.1 Product identification via the SICK product ID

SICK product ID

The SICK product ID uniquely identifies the product. It also serves as the address of the web page with information on the product.

The SICK product ID comprises the host name pid.sick.com, the part number (P/N), and the serial number (S/N), each separated by a forward slash.

For many products, the SICK product ID is displayed as text and QR code on the type label and/or on the packaging.



Figure 1: SICK product ID

3.1.2 Device view



Figure 2: Device view

- ① Aluminum housing with T-slot
- ② Front end cap with capacitive teachpad
- ③ 2 x LED indicators
- ④ Cable outlet
- ⑤ Rear end cap

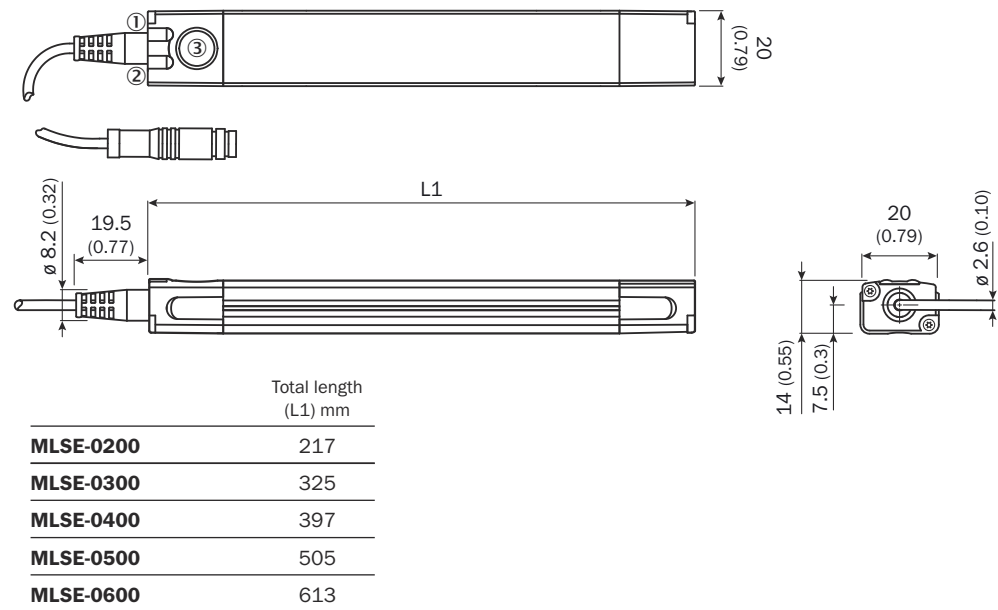


Figure 3: Dimensional drawing

3.2 Product characteristics

3.2.1 Product features

The MLS is a non-contact sensor used to determine the position of a magnetic track tape or the position of up to three track tapes simultaneously; these tapes are primarily used in conjunction with AGCs (autonomous guided carts).

The sensor is also capable of detecting defined markers positioned adjacent to the actual line.



NOTE

From firmware version 5, an inertial measuring unit (IMU) is integrated as an additional navigation aid which allows for detection of the current orientation of the sensor (see ["Inertial measuring unit", page 53](#)).

Other new features of firmware version 5 are:

- Bitmap, see ["Bitmap", page 44](#)
- Track optimization, see ["Track optimization", page 45](#)
- Temperature sensor, see ["Temperature sensor", page 57](#)
- Event detection, see ["Event detection", page 57](#)

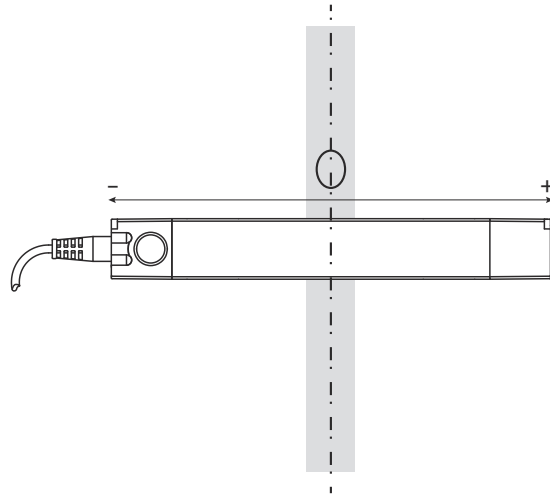
3.2.2 Operating principle

Principle of operation

The MLS determines the position of the magnetic tape via a row of hall sensors. The raw data from the hall sensors is fed into an algorithm to calculate the position of the line tape.

Line position

The line position is indicated relative to the geometric center of the longitudinal axis of the sensor. However, this only applies when the user has not set an offset for the zero position.



Resolution

Resolution refers to the minimum position change of the magnetic tape relative to the sensor that can be reflected in the sensor output.

Cycle time

The cycle time indicates the time interval within which the sensor can provide a new output signal.

Repeatability

Repeatability refers to the change in output signal when the relative position of the magnetic tape to the sensor does not change.

Orientation of the sensor

The orientation of the sensor determines the tilt of the sensor around the gravity axis (expressed with roll and pitch angle) as well as the torsion of the sensor around this axis (expressed by the yaw angle).

3.2.3 Suitable magnetic tape

The recommended magnetic tape width is 25 mm. Other dimensions should be tested for suitability before use.

The sensor is compatible with

- Magnetic tape with a single pole pair (north or south), unipolar magnetization on one side
- Various magnetic tape widths
- Various distances from the magnetic tape

Table 1: Magnetic tape



With magnetic tape that uses another form of magnetization (e.g., strip magnetization), correct functioning cannot be guaranteed.

Suitable magnetic tapes are available as accessories for the MLS, see www.sick.com/MLS.



NOTE

The suitability of foreign magnetic tapes should be checked before use. Use the MLS configuration software for this purpose (download via www.sick.com/MLS).

1. Hold the MLS over the magnetic tape at the desired working distance.
- ✓ The Track Level output in the configurator should be ≥ 4 and therefore in the green range.

Please note that magnetic tapes of inferior quality may experience a sharp reduction in field strength over their useful life and may therefore no longer be sufficient for navigation.

Table 2: Track Levels

	Track Level
Good (green) Sensor detects the tape as specified	7
	6
	5
	4
Critical (yellow) Sensor detects the tape as specified, but near the limits	3
	2
	1
Bad (red) Sensor detects the tape, but not within the specified range	0

4 Transport and storage

4.1 Transport

For your own safety, please read and observe the following notes:



NOTE

Damage to the sensor due to improper transport.

- The device must be packaged for transport with protection against shock and damp.
- Recommendation: Use the original packaging as it provides the best protection.
- Transport should be performed by specialist staff only.
- The utmost care and attention is required at all times during unloading and transportation on company premises.
- Note the symbols on the packaging.
- Do not remove packaging until immediately before you start mounting.

4.2 Transport inspection

Immediately upon receipt at the receiving work station, check the delivery for completeness and for any damage that may have occurred in transit. In the case of transit damage that is visible externally, proceed as follows:

- Do not accept the delivery or only do so conditionally.
- Note the scope of damage on the transport documents or on the transport company's delivery note.
- File a complaint.



NOTE

Complaints regarding defects should be filed as soon as these are detected. Damage claims are only valid before the applicable complaint deadlines.

4.3 Storage

Store the device under the following conditions:

- Recommendation: Use the original packaging.
- Do not store outdoors.
- Store in a dry area that is protected from dust.
- So that any residual damp can evaporate, do not package in airtight containers.
- Do not expose to any aggressive substances.
- Protect from sunlight.
- Avoid mechanical shocks.
- For storage periods of longer than 3 months, check the general condition of all components and packaging on a regular basis.

5 Mounting

5.1 Preparation for mounting

5.1.1 Mounting requirements



NOTICE

Radio interference may occur when the sensor is used in residential areas.

Only use the device in industrial environments (EN 61000-6-4).

- Typical space requirements for sensor, [see "Mechanics/electronics", page 62](#).
- Comply with technical data such as the permitted ambient conditions for operation of the sensor (e.g., temperature range, EM interference), [see "technical data", page 62](#).
- Protect the sensor from direct sunlight.
- Only affix the sensor using accessories supplied for this purpose -> there are no screw connections on the sensor.

Mounting location

When selecting the mounting location, the following factors must be considered:

- The mounting site must be as free from (electro-)magnetic interference fields as possible and should therefore not be in the immediate vicinity of unshielded electric motors.
- The sensor should not be mounted in the direct vicinity of ferromagnetic materials. If this cannot be avoided, the interference field compensation in [section 8.4.6](#) can help.
- The sensor must be fitted on the AGC at a 90° angle to the direction of travel.
- The sensor must be fitted with the greatest possible accuracy in a horizontal position.
- The sensor must be fitted with the label facing upward.
- The center point of the sensor should be positioned at the AGC center point, at a right angle to the direction of travel if possible.
- The working distance between the sensor and SICK magnetic tape is 10 mm - 70 mm¹⁾. The use of other magnetic tapes is possible, but should be tested in advance ([see "Suitable magnetic tape", page 11](#)).

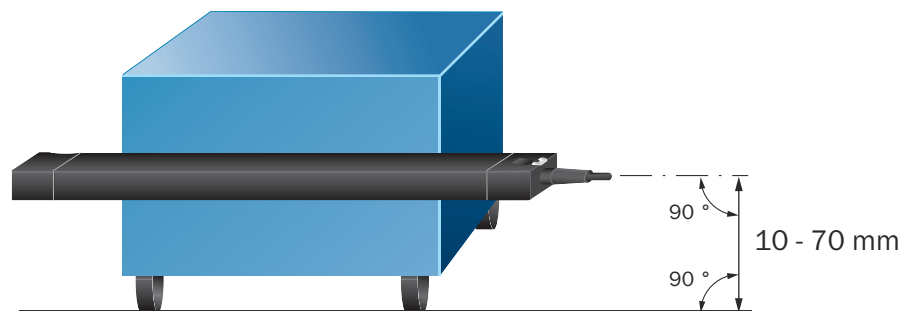


Figure 4: Mounting location

5.1.2 Scope of delivery

The following are included with delivery:

¹⁾ with magnetic averaging value "4", [see "Average filter for track output values", page 48](#).
The working distance with the previous firmware version V2 is 10-50 mm.

- 1 MLS sensor in the version ordered
- 1 quick-start guide

Accessories:

Accessories (e.g., cables, fastening adapters) are only supplied if ordered separately.

The available accessories are listed at www.sick.com/mls.

5.2 Mounting the sensor

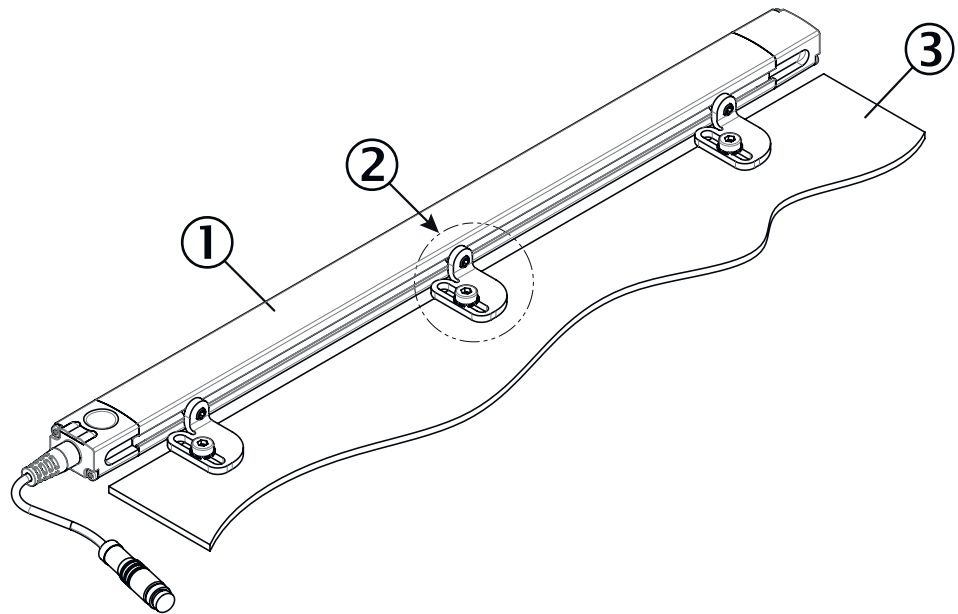


Figure 5: Horizontal mounting

- ① MLS-xxx
- ② Mounting bracket, part number number 2065973
- ③ Mounting panel

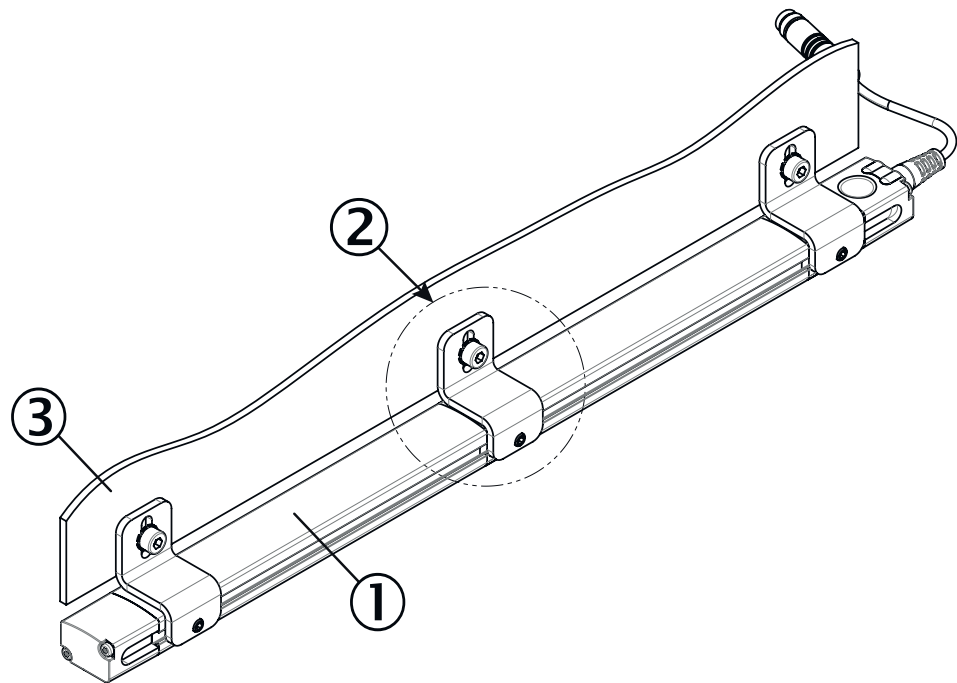


Figure 6: Vertical mounting

- ① MLS-xxx
- ② Mounting bracket, part number number 2065577
- ③ Mounting panel

5.3 Mounting of the magnetic tape

To navigate AGCs, the MLS is aligned to the Line Center Point of a magnetic tape, which specifies the route of the AGC. The magnetic tape is stuck to the ground or embedded in the ground. The criteria for selecting the suitable magnetic tape are described in chapter 3.2.3. SICK offers respective self-adhesive magnetic tapes as accessories for the MLS (see www.sick.com/MLS).



NOTE

The following information is based on internal test results. However, this does not mean that all users do not have to test the suitability of the magnetic tape for their intended purposes.

5.3.1 Preparation

The SICK magnetic tape is well-suited for laying in the inner area. The following points must be ensured before mounting the magnetic tape:

- The surface must be fixed, flat and closed.
 - Moisture and contamination such as dirt, grease, dust, etc. must be removed to make sure the magnetic tape will really hold. Use clean cloths and solvents (e.g. benzine, acetone, pure alcohol) or floor cleaners with degreasing effect. Make sure the substrate can tolerate any solvents used.
- Any paint surfaces must adhere firmly, be free of solvents and silicone and be completely dry.
- After cleaning with solvents, let the substrate air out for about 10 minutes.

- The ambient and surface temperature should be at least +15 °C; if applicable, provide for supply of warm air before, during and after processing. This activates the adhesive very well.
- The route should be as free of ferromagnetic materials as possible.

A sample adhesion of the magnetic tape is recommended.

5.3.2 Laying

Before laying the magnetic tape, plan out the route on the ground in line with the layout. Gaps and magnetic tape overlap should be avoided. Overlaps can be used specifically for intersections (see Intersections).

- Remove about 10 cm of protective paper from the back of the magnetic tape.
- Place the exposed end of the strip in the correct position. Press firmly. Use your hand to peel off the rest of the protective paper. Rub the strip well with the other hand. With a suitable pressure roller or a wood roller, press down on the marking tape.
- With wide strips, always roll from the center to the edge to roll out air pockets.

Curves

The magnetic tapes from SICK are suitable for laying curves with large radii. Smaller radii can be laid with the curved magnetic tape available as an accessory. Alternatively, the magnetic tape can be cut and stuck together in a flush position. A minimum curve radius of 1.5 m must be ensured; also ensure that the “field of view” of the MLS on the AGC does not leave the track, which causes detection to fail.

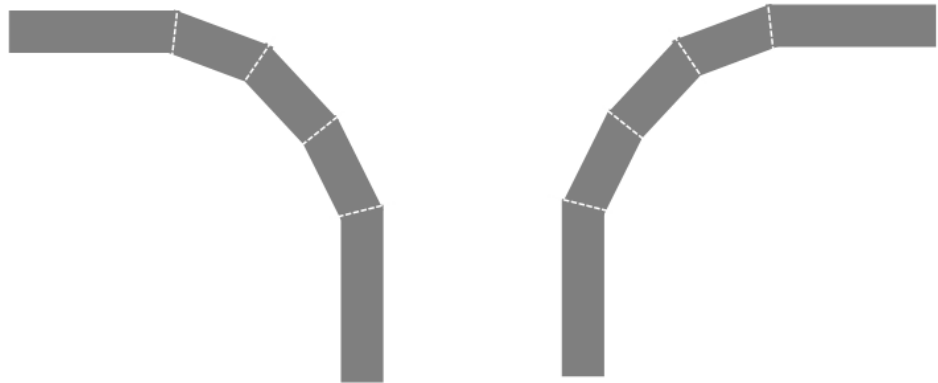
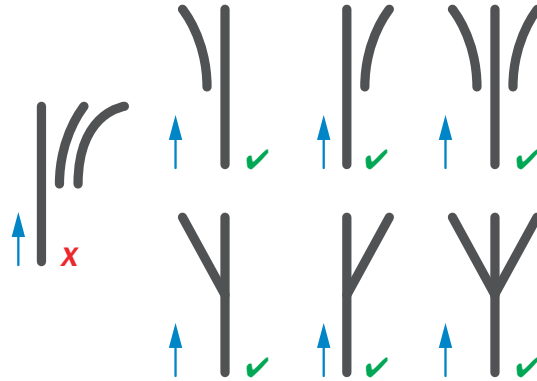


Figure 7: Laying curves

Diverter

To create diverters, one track per side can be detected in addition to the main track. If the diverters are stuck together so that the forked track comes directly out of the main track, data output as **improved track detection at diverters** is recommended (can be adjusted using Index 0x2006, Subindex 1, [see "Process data objects", page 36](#)).



Diverter, in the case of the flush variant, should not be overlapping. In line with this, the area marked in red should always be removed.

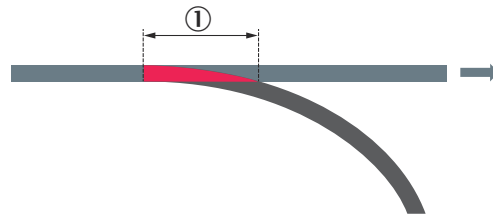


Figure 8: Flush variant

① 150 ... 200 mm

With the non-flush variant, for the best possible detection, the diverter should run 150 mm parallel to the main track before changing direction. The distance to the main track should not be less than the track width.

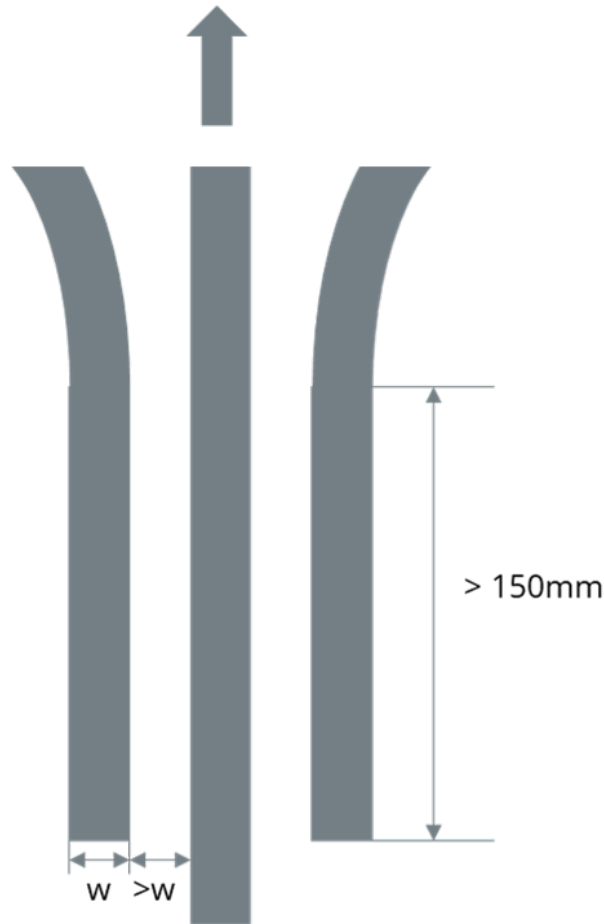


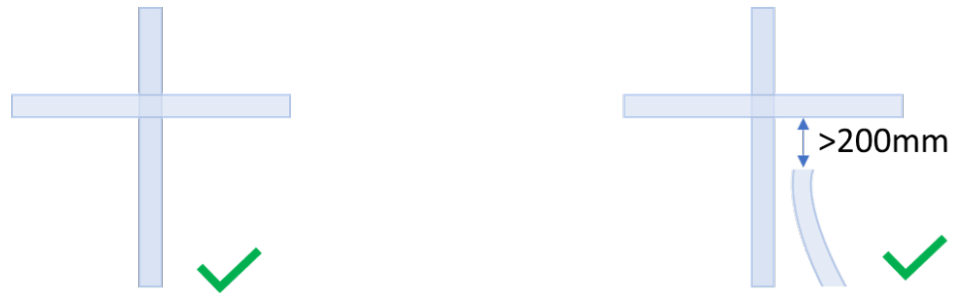
Figure 9: Non-flush variant

Intersections

For track intersections (the track tapes cross at a 90° angle), one track should be completely glued over the other. No diverter should follow at a distance of 200 mm before and after the intersection, taking into account the speed of travel.

Splitting a track tape and attaching a bump to the through track tape is not recommended. This leads to difficulties in track detection, since the sensor does not output a valid position value in the intersection. Only in the ideal case does the sensor detect such a situation and output the following process data:

#LCP = 7; LCP1 = LCP2 = LCP3



Zones

The MLS can detect and differentiate between both magnetic tape with north pole on the top side (5337613) and south pole on the top side (5337614). To limit zones on a route, it is therefore possible to change from one magnetic tape to the other in a targeted manner.

Markers

The MLS is capable of detecting magnetic codes using so-called markers next to the main track. For a detailed description, see chapter [section 8.5](#).

For more information on parameterization, see [section 8.4.6](#).

5.3.3 Protection

To protect the magnetic tape if there is a strong mechanical load, a protective tape or protective epoxy layer can be attached to the magnetic tape.

In general, rotating or turning vehicles on the magnetic tape, as well as moving objects like pallets against the tape, should be avoided. If this cannot be avoided, lowering the magnetic tape into the ground is recommended.

6 Electrical installation

6.1 Safety

6.1.1 Notes on the electrical installation



CAUTION

Danger due to incorrect supply voltage!

An incorrect supply voltage may result in injuries from electric shocks and/or damage to the device.

- Only operate the sensor with safety extra-low voltage (SELV).



NOTICE

Sensor damage or unpredictable operation due to working with live parts.

Working with live parts may result in unpredictable operation.

- Only carry out wiring work when the power is off.
- Only connect and disconnect electrical connections when the power is off.

- **The electrical installation must only be performed by electrically qualified personnel.**
- **Standard safety requirements must be met when working in electrical systems.**
- Only switch on the supply voltage for the device when the connection tasks have been completed and the wiring has been thoroughly checked.
- When using extension cables with open ends, ensure that bare wire ends do not come into contact with each other (risk of short-circuit when supply voltage is switched on!). Wires must be appropriately insulated from each other.
- Wire cross-sections in the supply cable from the customer's power system must be designed in accordance with the applicable standards. When this is being done in Germany, observe the following standards: DIN VDE 0100 (Part 430) and DIN VDE 0298 (Part 4) and/or DIN VDE 0891 (Part 1).
- Circuits connected to the device must be designed as SELV circuits (SELV = Safety Extra Low Voltage).
- Protect the device with a separate fuse at the start of the supply circuit.

A shielded cable is not required in order to adhere to the electromagnetic compatibility guidelines specified by EN 61000-6-2/4. It is recommended, however, especially when working with longer connecting cables.

The IP enclosure rating for the sensor is only achieved if the connected cable is completely screwed in.

6.1.2 Wiring notes



NOTE

Preassembled cables can be found online at:

Please observe the following wiring notes:

- During installation, pay attention to the different cable groups. The cables are grouped into the following four groups according to their sensitivity to interference or radiated emissions:

- Group 1: Cables very sensitive to interference, such as analog measuring cables
- Group 2: Cables sensitive to interference, such as sensor cables, communication signals, bus signals
- Group 3: Cables which are a source of interference, such as control cables for inductive loads, motor brakes
- Group 4: Cables which are powerful sources of interference, such as output cables from frequency inverters, welding system power supplies, power cables
- ▶ Cables in groups 1, 2 and 3, 4 must be crossed at right angles, [see figure 10](#).
- ▶ Cables in groups 1, 2 and 3, 4 must be routed in different cable channels or metallic separators must be used, [see figure 11](#) and [see figure 12](#). This applies particularly where cables of devices with a high level of radiated emission, such as frequency converters, are laid parallel to sensor cables.

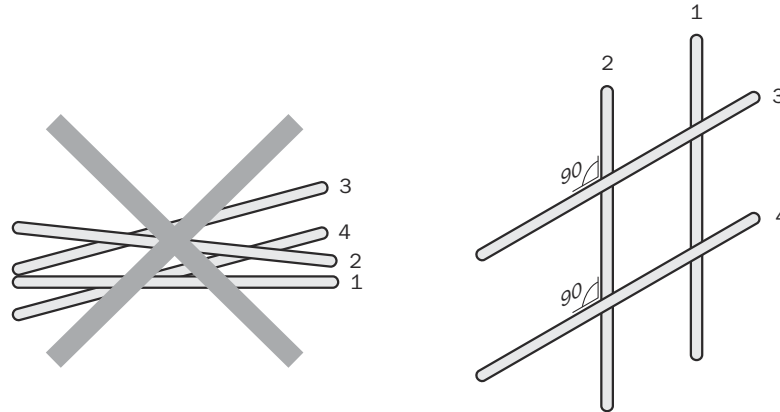


Figure 10: Cross cables at right angles

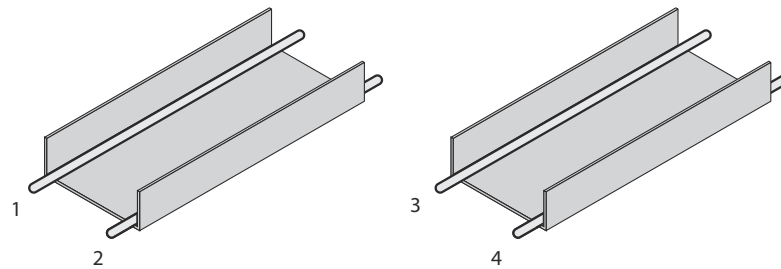


Figure 11: Ideal laying – Place cables in different cable channels

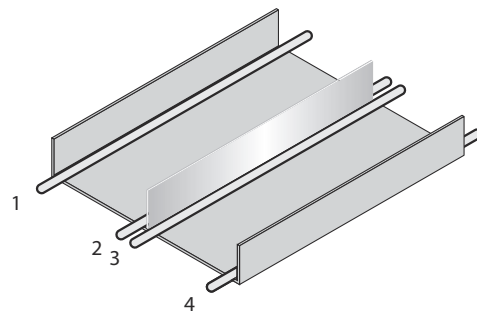


Figure 12: Alternative laying – Separate cables with metallic separators



NOTE

Prevent equipotential bonding currents via the cable shield with a suitable grounding method, [see "Safety", page 21](#).

6.2 Pin assignment of the connections

M8 connection

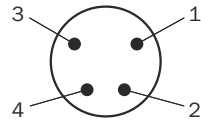


Figure 13: Pin assignment: M8-male, A-coded, 4-pin

Pin	Wire color	Pin assignment
1	Brown	L+
2	White	CAN-HIGH
3	Blue	GND
4	Black	CAN-LOW

M12 connection

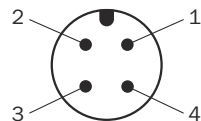


Figure 14: Pin assignment: M12-male, A-coded, 4-pin

Pin	Wire color	Pin assignment
1	Brown	L+
2	White	CAN-HIGH
3	Blue	GND
4	Black	CAN-LOW

Cable outlet

Wire color	Pin assignment
Brown	L+
White	CAN-HIGH
Blue	GND
Black	CAN-LOW

6.3 Connecting the supply voltage

The sensor must be connected to a voltage supply with the following properties:

- Supply voltage DC 9 V ... 30 V (stabilized safety extra-low voltage [SELV] as per current standard EN 60950-1)
- Electricity source with at least 5 W power

Protecting the supply cables

To ensure protection against short-circuits/overload in the customer's supply cables, the conductor cross sections used must be appropriately selected and protected.

The following standards must be observed in Germany:

- DIN VDE 0100 (part 430)
- DIN VDE 0298 (part 4) and/or DIN VDE 0891 (part 1)

6.4 CAN connection

We recommend connecting the ground cable to the CANopen communication interface as a reference.

The sensor is available with a 120 Ohm terminator as an option.

The selection of the version with or without the internal terminator is made via the product short text:

- MLSE-0200A2TP0 → T = Termination, internal 120 Ohm terminator
- MLSE-0200A2NP0 → N = No termination, no internal 120 Ohm terminator



NOTE

For the variant without an internal terminator, we strongly recommend connecting an external 120 Ohm terminator at both ends of the CANopen bus network for maximum protection against electromagnetic interference.

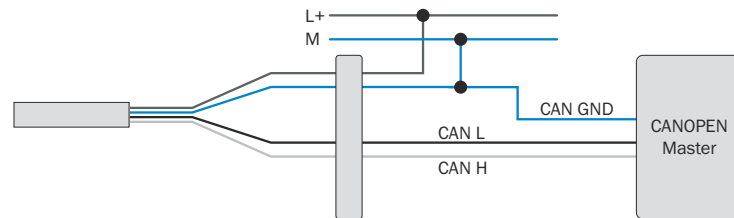


Figure 15: MLS with terminator

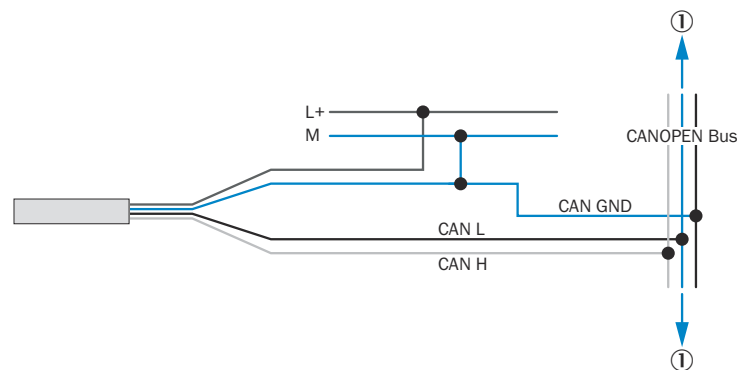


Figure 16: MLS without terminator

7 Commissioning

7.1 Overview of commissioning steps

- Connect the voltage supply.
- Commission the sensor using the factory settings.
- Configure the sensor.

7.2 Commissioning the sensor for the first time

Establish voltage supply: When the voltage supply is correct and no line is detected by the sensor, the green LED flashes in the following pattern:

ON: 0.8 s, OFF: 0.2 s

If at least one line is detected with sufficient magnetic field strength, the yellow LED lights up.

If exactly one line is detected, the approximate position is indicated via the green and blue LED.

7.3 Quick-start guide for the MLS CANopen

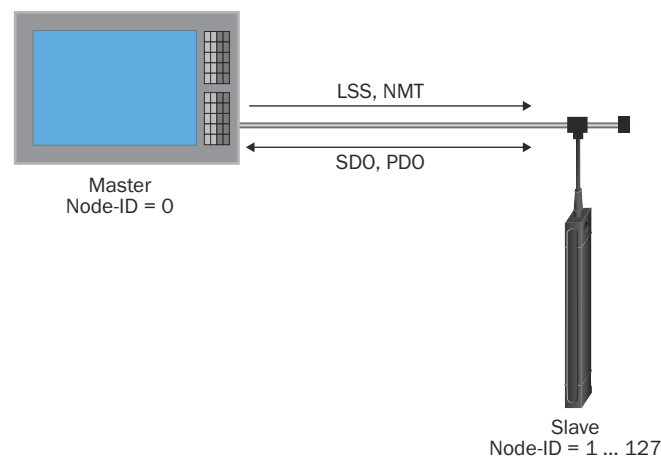
7.3.1 Setting ID and bit rate

The following conditions must be met for communication with the master:

- A correct node ID must be set on the MLS.
The following are correct:
 - A node ID which is free in the CANopen network
 - A node ID which the master expects
- The same bit rate must be set on the MLS as in the master.

The following parameters are factory set on the MLS:

- Node ID: 0x0A
- Bit rate: 125 kbit
- TPD01 active: Time-controlled, 10 ms cycle time



The following communication parameters can be allocated to the MLS:

- Node ID: 1 to 127 (0 is generally assigned to the master)
- Bit rate: 10 kbit/s, 20 kbit/s, 50 kbit/s, 100 kbit/s, 125 kbit/s, 250 kbit/s, 500 kbit/s, 800 kbit/s, 1,000 kbit/s

**NOTE**

The bus load increases with low transmission rates.

The node ID and bit rate are set as follows:

Using layer setting services (See also: CiA Standard 305: “Layer setting services (LSS) and protocols”).

**NOTE**

After amending the node ID, the **Reset Communication** command (NMT command 0x82) must be executed.

The NMT command is always transmitted with COB-ID 0x000 and in the following format:

Table 3: COB-ID 0x000

COB-ID	rtr	len	NMT function	Target node
0x000	0	2	1 byte	1 byte

The target node is either a specific node 0x01-0x7F or 0x00 for all nodes in the bus network.

After amending the bit rate, the sensor must be restarted.

The node ID and bit rate can also be set via the MLS configuration software (download via www.sick.com/MLS).

1. Connect the MLS to your PC (you can find matching cables and adapters online www.sick.com/MLS) and start the configuration software.

**NOTE**

Installing the driver is required for use of the CAN USB converter.

2. Establish a connection between the sensor and configurator via **Device – Connect**.
3. In the connection settings, enter the desired bit rate and NodeID and confirm with **OK**. The entered settings are saved in the sensor.

**NOTE**

The saved bit rate and NodeID cannot be read out via the configurator. It is ideal to note the saved settings on the sensor.

7.3.2 Process data objects (PDOs)

The MLS supports up to seven transmit PDOs (up to four at a time) and no receive PDOs.

Objects 1800h to 1806h contain the communication parameters. The mapping is fixed and cannot be changed.

7.3.3 PDO communication

The transmission type is factory-set to time driven for TPDO1 in object 1800h. This means that the sensor sends its TPDO cyclically. Only TPDO1 is activated at the factory, other TPDOs (TPDO2 in firmware V2 and TPDO2-TPDO7 in firmware V5) must be activated via the COB-ID before use.

Changing the factory-set transmission type

The following options are available for the cyclical or acyclical output of transmit-PDOs by the MLS:

- Change the event timer in object 1800h ([see table 4](#))
- Change the transmission type in object 1800h ([see "Transmission types", page 40](#))

The same adjustments apply to TPD02 to TPD07 in indexes 1801h to 1806h.

Take note of the inhibition time**NOTE**

The inhibition time of the PDOs limits the communication of a device to the CANopen bus. It always has higher priority than the event timer, the CoS events, and Sync triggering.

If, for example, the event timer is set to 100 ms and the inhibition time to 1 s, the respective PDO is only sent every second.

If the event timer is set to below 10 ms, a new TPD0 may be transmitted before any new data has been recorded by the sensor. In such cases, the old data is transmitted again.

8 Operation

8.1 CANopen

All sensor functions can be accessed via the CANopen interface. All settings can be configured in this interface.

The EDS file can also be found at www.sick.com/mls

8.1.1 CANopen service data object directory

This section only describes the SDOs that do not have a fixed definition in the CANopen standard. Default values are listed only for parameters that can be modified by the user.



NOTE

The SDOs marked with a footnote are implemented from firmware version V5.

Table 4: SDOs

Index	Sub	R/W	Data type	Object name	Default value	Description
0x1000		RO	UINT32	Device type	-	No device profile supported
0x1001		RO	UINT8	Error register	-	No error output
0x1008		RO	STRING	Manufacturer device name	-	Magnetic line guidance sensor
0x1009		RO	STRING	Manufacturer hardware version	-	Hardware version, sensor
0x100A		RO	STRING	Manufacturer software version	-	Firmware version, sensor
0x100C ¹		R/W	UINT16	Guard time	0	Node guarding not supported
0x100D ¹		R/W	UINT8	Life time factor	0	Node guarding not supported
0x1010 ¹				Store parameter field		
	1	R/W	UINT32	Save all parameters	2	Sets the saving behavior of the CANopen parameters
0x1017 ¹		R/W	UINT16	Heartbeat time	0	Heartbeat time in [ms]
0x1018				Identity object		
	1	RO	UINT32	Vendor-ID	-	0x01000056 (SICK AG)
	2	RO	UINT32	Product code	-	0x00001100 (MLS)
	3	RO	UINT32	Revision number	-	0x00000005

Index	Sub	R/W	Data type	Object name	Default value	Description
	4	RO	UINT32	Serial number	-	Format: YYWW-nnnn YY: Year of production WW: Week of production nnnn: Sequential number
0x1200 ¹				Server SDO parameter		
	1	RO	UINT32	COB ID client to server	-	Standard COB-ID for SDO transfer master to sensor
	2	RO	UINT32	COB ID server to client	-	Standard COB-ID for SDO transfer sensor to master
0x1800				Transmit PDO communication parameter 0		
	1	R/W	UINT32	COB ID	0x0000018A	see "Transmission types", page 40
	2	R/W	UINT8	Transmission type	0xFF	see "Transmission types", page 40
	5	R/W	UINT16	Event timer	0x000A	see "Transmission types", page 40
0x1801 ¹				Transmit PDO communication parameter 1		
	1	R/W	UINT32	COB ID	0x8000028A	see "Transmission types", page 40
	2	R/W	UINT8	Transmission		see "Transmission types", page 40
	5	R/W	UINT16	Event timer		see "Transmission types", page 40
0x1802 ¹				Transmit PDO communication parameter 2		
	1	R/W	UINT32	COB ID	0x8000038A	see "Transmission types", page 40
	2	R/W	UINT8	Transmission type	0xFF	see "Transmission types", page 40
	5	R/W	UINT16	Event timer	0x0000	see "Transmission types", page 40

Index	Sub	R/W	Data type	Object name	Default value	Description
0x1803 ¹				Transmit PDO communication parameter 3		
	1	R/W	UINT32	COB ID	0x8000048A	see "Transmission types", page 40
	2	R/W	UINT8	Transmission type	0xFF	see "Transmission types", page 40
	5	R/W	UINT16	Event timer	0x0000	see "Transmission types", page 40
0x1804 ¹				Transmit PDO communication parameter 4		
	1	R/W	UINT32	COB ID	0x80000000	see "Transmission types", page 40
	2	R/W	UINT8	Transmission type	0xFF	see "Transmission types", page 40
	5	R/W	UINT16	Event timer	0x0000	see "Transmission types", page 40
0x1805 ¹				Transmit PDO communication parameter 5		
	1	R/W	UINT32	COB ID	0x80000000	see "Transmission types", page 40
	2	R/W	UINT8	Transmission type	0xFF	see "Transmission types", page 40
	5	R/W	UINT16	Event timer	0x0000	see "Transmission types", page 40
0x1806 ¹				Transmit PDO communication parameter 6		
	1	R/W	UINT32	COB ID	0x80000000	see "Transmission types", page 40
	2	R/W	UINT8	Transmission type	0xFF	see "Transmission types", page 40
	5	R/W	UINT16	Event timer	0x0000	see "Transmission types", page 40 ¹
0x1F80 ¹		R/W	UINT32	NMTStartup	8	Sets the start-up behavior of the MLS
0x2000 ¹				Hall values		

Index	Sub	R/W	Data type	Object name	Default value	Description
	1-A8	RO	INT16		-	Raw values of the sensor elements
0x2006				Variant configuration		
	1	R/W	UINT8	Variant TPD01	3	Switching data output in TPD01, see "Process data objects", page 36  NOTE After a change, the device must be restarted for activation.
	2	R/W	UINT8	Variant MEMS	1	De-/Activation of the integrated IMU including temperature sensor, see "Inertial measuring unit", page 53  NOTE After a change, the device must be restarted for activation.
0x2019		RO	UINT32	Order number	-	Part number of the sensor
0x2021				LCPs		see "Process data objects", page 36
	1	RO	UINT16	LCP1	-	Line Center Point 1
	2	RO	UINT16	LCP2	-	Line Center Point 2
	3	RO	UINT16	LCP3	-	Line Center Point 3
	4	RO	UINT8	#LCP	-	Figure for number of detected tracks and markers
	5 ¹	RO	UINT8	LCP1_width	-	Width of track tape 1
	6 ¹	RO	UINT8	LCP2_width	-	Width of track tape 2
	7 ¹	RO	UINT8	LCP3_width	-	Width of track tape 3
	8 ¹	RO	UINT16	LCP1_Combi	-	Combination of LCP1 and LCP1_width

Index	Sub	R/W	Data type	Object name	Default value	Description
	9 ¹	RO	UINT16	LCP2_Combi	-	Combination of LCP2 and LCP2_width
	A ¹	RO	UINT16	LCP3_Combi	-	Combination of LCP3 and LCP3_width
	B ¹	RO	INT8	Line Level 1	-	Measured magnetic field strength of each individual track (positive or negative) LSB = 0.048mT
	C ¹	RO	INT8	Line Level 2	-	
	D ¹	RO	INT8	Line Level 3	-	
0x2022		RO	UINT8	Status	-	see "Process data objects", page 36
0x2023		RO	UINT16	Track Level	-	see table 18, page 43
0x2024		RO	UINT16	Field level	-	Measured magnetic field strength with full resolution
0x2025		R/W	UINT16	Min. level	200	Minimum magnetic field strength in digits above which a line is detected
0x2026		R/W	INT16	Offset	0	Offset [mm] for sensor zero point
0x2027		R/W	BOOL	Sensor flipped	0	Sensor rotated by 180° 0 => Positive positions on cable outlet 1 => Negative positions on cable outlet
0x2028				Markers		see "Marker detection", page 51
	1	R/W	BOOL	Use markers	0	0 => No marker detected in TPDO 1 => Detected markers displayed in TPDO1
	2	R/W	UINT8	Marker style	0	0 => No marker detection 1 => SICK marker standard mode 2 => SICK marker extended mode see "Marker detection", page 51

Index	Sub	R/W	Data type	Object name	Default value	Description
	3	R/W	BOOL	Failsafe mode	0	0 => Failsafe mode deactivated 1 => Failsafe mode activated see "Marker detection", page 51
0x2029		R/W	BOOL	Lock teach	0	0 => Teach key activated 1 => Teach key deactivated
0x202A		WO	BOOL	Set param. to default	-	1 => All parameters are set to factory defaults  NOTE To reset the parameters "Transmission type" and "Event timer", a communication reset must also be performed.
0x202B		WO	BOOL	Trigger user offset calibration	-	1 => Offset calibration of the hall elements is performed; the sensor is then restarted
0x202C		WO	BOOL	Trigger zero position teach	-	1 => Teaches the current position as the new zero point
0x202D ¹				Detection config		
	1	R/W	UINT8	First Level	30	Lowest Detection Level for track detection in [%] of the main peak
	2	R/W	UINT8	Last level	30	Highest Detection Level for track detection in [%] of the main peak
	3	R/W	UINT8	Averaging magnetic	1	Average filter for track output values
	4	R/W	UINT8	Averaging inertial	0	Average filter for orientation, see "Inertial measuring unit", see "Inertial measuring unit", page 53

Index	Sub	R/W	Data type	Object name	Default value	Description
	5	R/W	UINT8	Tape polarity	0	Track polarity 0 - both polarities valid 1 - track has north polarity 2 - track has south polarity
0x202E ¹				Bitmap		see "Bitmap", page 44
	1	RO	UINT32	Bitmap byte 1-4	-	Elements 1-32
	2	RO	UINT32	Bitmap byte 5-8	-	Elements 33-64
	3	RO	UINT32	Bitmap byte 9-12	-	Elements 65-96
	4	RO	UINT32	Bitmap byte 13-16	-	Elements 97-128
0x202F ¹				Bitmap config		Bitmap configuration
	1	R/W	UINT16	Bitmap threshold	500	Detection threshold
	2	R/W	UINT8	Bitmap mode	0	0 - deactivated 1 - both polarities valid 2 - only south polarity valid 3 - only north polarity valid
0x2030 ¹				Orientation of Euler angles		Orientation in Euler angle representation in 1/10000rad,
	1	RO	INT16	Roll	-	Roll angle ϕ
	2	RO	INT16	Pitch	-	Pitch angle θ
	3	RO	INT16	Yaw	-	Yaw angle ψ
0x2031 ¹				Orientation of quaternion		Orientation in quaternion display in 1/10000 (no unit), see "Inertial measuring unit", page 53
	1	RO	INT16	w (real part)	-	Real part
	2	RO	INT16	x (imaginary part)	-	1. Imaginary part
	3	RO	INT16	y (imaginary part)	-	2. Imaginary part
	4	RO	INT16	z (imaginary part)	-	3. Imaginary part
0x2032 ¹				IMU settings		see "Inertial measuring unit", page 53
	1	WO	UINT8	Reset yaw	-	Resetting yaw angle to 0
	2	WO	UINT8	Reset acceleration peaks	-	Resetting acceleration peaks

Index	Sub	R/W	Data type	Object name	Default value	Description
0x2033 ¹				Acceleration		Acceleration raw data in 2^{-11} g or 4.79×10^{-3} m/s ² , see "Inertial measuring unit", page 53
	1	RO	INT16	x-axis	-	x-axis
	2	RO	INT16	y-axis	-	y-axis
	3	RO	INT16	z-axis	-	z-axis
0x2034 ¹				Angular velocity		Yaw rate raw data in 125×2^{-11} °/s or 1.065×10^{-3} rad/s, see "Inertial measuring unit", page 53
	1	RO	INT16	x-axis	-	x-axis
	2	RO	INT16	y-axis	-	y-axis
	3	RO	INT16	z-axis	-	z-axis
0x2035 ¹		RO	UINT16	Sample time stamp	-	Time stamp (running over) in ms for raw data values / orientation, see "Inertial measuring unit", page 53
0x2070 ¹				Temperature		Temperature in °C
	1	RO	INT8	Current temperature	-	Current temperature
	2	RO	INT8	Max. temperature all time	-	Maximum temperature throughout entire operating time of the sensor
	3	RO	INT8	Min. temperature all time	-	Minimum temperature throughout entire operating time of the sensor
	4	R/W	INT8	Max. temperature since last reset	-	Maximum temperature since last reset
	5	R/W	INT8	Min. temperature since last reset	-	Minimum temperature since last reset
	6	R/W	INT8	Upper temperature threshold	80	Temperature threshold above which a temperature event is generated, see "Event detection", page 57

Index	Sub	R/W	Data type	Object name	Default value	Description
	7	R/W	INT8	Lower temperature threshold	-30	Temperature threshold below which a temperature event is generated, see "Event detection", page 57
2x2080 ¹				Event code		Error code (set when exceeding the temperature threshold, among other things), see "Event detection", page 57
	1	WO	UINT8	Reset event flag	-	Reset event code
	2	RO	UINT8	Event source	-	Event source
	4	RO	UINT8	Magnetic field	-	Magnetic field event code
	5	RO	UINT8	Temperature	-	Temperature event code

¹ This index is implemented from firmware version V5

8.1.2 Process data objects

TPDO activation

From firmware version 5, a total of 7 possible TPDOs are defined, of which a maximum of 4 can be activated at one time. The TPDOs with fixed mapping are activated/deactivated via the setting of the respective COB-ID (subindex 1):

Table 5: Overview of TPDOs

TPDO communication parameter	SDO Index	Default COB-ID (Sub-index 1)	Description
TPDO1	0x1800	0x180 + {NodeID}	Track position and width, marker, status
TPDO2	0x1801	0x80000280 + {NodeID}	Line Width and Line Level
TPDO3 ¹	0x1802	0x80000380 + {NodeID}	Bitmap
TPDO4 ¹	0x1803	0x80000480 + {NodeID}	Euler angle and time stamp
TPDO5 ¹	0x1804	0x80000000 + {NodeID}	Quaternion and time stamp
TPDO6 ¹	0x1805	0x80000000 + {NodeID}	Acceleration raw data and time stamp
TPDO7 ¹	0x1806	0x80000000 + {NodeID}	Rotation rate raw data and time stamp

¹ This index is implemented from firmware version V5.

In general: For a COB-ID, the MSB is set, so this PDO is deactivated. By default, TPDO2...TPDO7 are therefore deactivated. To activate a PDO, a valid COB-ID is assigned to it. The valid COB-IDs are 0x180, 0x280, 0x380 and 0x480 (each +{NodeID}). The COB-IDs can be freely assigned to the TPDOs. If, for example, in Index 1804, Subindex 1, the COB-ID 0x280+{NodeID} is written, the quaternions are sent as TPDO2.

To send the TPDO in **Time Driven** mode, value 255 must be entered in Subindex 2 for the communication parameter (default setting). First, the time interval in which the TPDO is sent must be set in [ms] in Subindex 5.

Description of TPDOs

The **TPDO1** contains track position and width, marker and status. By default, the TPDO1 is structured as follows:

Table 6: TPDO1

	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
TPDO1	LSB LCP1	MSB LCP1	LSB LCP2	MSB LCP2	LSB LCP3	MSB LCP3	#LCP	Status

Value	Data type	Description
LCP1	INT16	Position of Line Center Point 1 [mm]
LCP2	INT16	Position of Line Center Point 2 [mm]
LCP3	INT16	Position of Line Center Point 3 [mm]
#LCP	-	See table #LCP
Status	-	See status table

Table 7: TPDO1, byte 7: #LCP

	Bit 0	Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7
#LCP	#LCP Bit 0	#LCP Bit 1	#LCP Bit 2	Marker Bit 0	Marker Bit 1	Marker Bit 2	Marker Bit 3	Marker Bit 4

Value	Data type	Description
#LCP	UINT3	The following assignment applies (see "Output of line center points", page 42): 0 => No track found 2 => One track found 3 => Two tracks found: Left diverter 6 => Two tracks found: Left diverter 7 => Three tracks found or 90 °C intersection ²
Marker	INT5	Bit 0 is the introductory character bit Bit 1...4 represent code 1...15

² see "Laying", page 17

Table 8: TPDO1, byte 8: Status

	Bit 0	Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7
Status	Line good	Track Level Bit 0	Track Level Bit 1	Track Level Bit 2	Sensor flipped	Polarity	Reading code	Event flag

Value	Data type	Description
Line good	BOOL	0 => No track or track too weak 1 => Sufficiently strong track detected The status depends on the conditions of the FieldLevel . The hysteresis of the FieldLevel is defined as 400 ... 500. When a line is detected and the FieldLevel reaches a value greater than 500, the Line good status is activated. If the line disappears, the Line good status is retained until the FieldLevel falls below 400. Only then is the Line good status reset (value = "0").

Value	Data type	Description																								
Track Level	UINT3	Display of magnetic field strength in accordance with the following table: Table 9: Track Level <table><tr><th>FieldLevel</th><th>Line Level</th></tr><tr><td>4500</td><td>7</td></tr><tr><td>2250</td><td>7</td></tr><tr><td>2000</td><td>7</td></tr><tr><td>1750</td><td>6</td></tr><tr><td>1500</td><td>5</td></tr><tr><td>1250</td><td>4</td></tr><tr><td>1000</td><td>3</td></tr><tr><td>750</td><td>2</td></tr><tr><td>500</td><td>1</td></tr><tr><td>450</td><td>0</td></tr><tr><td>350</td><td>0</td></tr></table>  NOTE Detailed table see table 19, page 43.	FieldLevel	Line Level	4500	7	2250	7	2000	7	1750	6	1500	5	1250	4	1000	3	750	2	500	1	450	0	350	0
FieldLevel	Line Level																									
4500	7																									
2250	7																									
2000	7																									
1750	6																									
1500	5																									
1250	4																									
1000	3																									
750	2																									
500	1																									
450	0																									
350	0																									
Sensor flipped	BOOL	Indicates whether or not the measuring range has been inverted 0 => Negative positions on cable connection side 1 => Positive positions on cable connection side																								
Polarity	BOOL	Indicates whether the upper surface of the magnetic tape is magnetized to the north or south pole 0 => North pole 1 => South pole																								
Reading code	BOOL	0 => No code present to read 1 => Sensor is reading code																								

Setting options of the TPDO1:

The track data, the interference field suppression and the improved diverter detection can be parameterized with the TPDO1.

The track data in the TPDO1 can be output with and without track width. The maximum value of the track width that can be output in TPDO1 is 63 mm (= 6 bit). If the track data type is set to "Standard", the track data is output in TPDO1 without track width or as shown in Table 7. If track data type "Combi" is set, the track data is output with the track width. When the track data type is changed, a voltage reset is necessary to activate the change. Note: The track width can also be output via the TPDO2. The maximum value is not limited here (16 bit). However, this additional TPDO naturally increases the bus load, whereas selecting track data type "Combi" does not increase the bus load.

The interference field suppression can blank magnetic interference fields such as offset fields caused by magnetized steel parts in a concrete floor.

Improved diverter detection allows for smooth navigation when diverters are laid flush.

The following table selection describes the setting options of the TPDO1 Index (0x2006 Sub1):

Table 10: Interference field suppression

Value	Name	Track data	Interference field suppression	Improved diverter detection
0	Standard	Standard	-	-
1	Combi	Combi	-	-
2	Standard compensated	Standard	x	-
3	Standard enhanced	Standard	-	x
4	Standard enhanced compensated	Standard	x	x
5	Combi compensated	Combi	x	-
6	Combi enhanced	Combi	-	x
7	Combi enhanced compensated	Combi	x	x

For example, this means that the TPDO1 should be parameterized to the value 4 in an environment with magnetic interference fields in the floor, flush diverters and a normal bus load, with only the requirement of the line center point.

If the diverters are not laid flush and a track width is required, the bus load is already increased and no interference fields occur, the TPDO1 can be parameterized to 1.

An alternative output of data is offered by value 1 in Index 0x2006, Subindex 1. The information on line center point (LCP) and line width (LCP_width) are combined in one value here (2 byte). The changed data types are to be observed here:

- LCPx: INT10 (value range -511 ... 511)
- LCPx_width: UINT6 (value range 0 ... 63)

The two bytes are divided up as follows:

Byte	Bit 0	Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7
LSB	LCPx (0 ... 7)							
MSB	LCPx (8) LCP_width (0 ... 5)							LCPx (9)

TPDO2 contains Line Width and Line Level:

Table 11: TPDO2

	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
TPDO2	Width LCP1	Width LCP2	Width LCP3	Line Level LCP1	Line Level LCP2	Line Level LCP3		

TPDO3 contains the bitmap:

Table 12: TPDO3

	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
TPDO3	Bitmap byte 1	Bitmap byte 2	Bitmap byte 3	Bitmap byte 4	Bitmap byte 5	Bitmap byte 6	Bitmap byte 7	Bitmap byte 8

TPDO4 contains the orientation in Euler angles and time stamps:

Table 13: TPDO4

	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
TPDO4	LSB roll	MSB roll	LSB pitch	MSB pitch	LSB yaw	MSB yaw	LSB time stamp	MSB time stamp

TPDO5 contains the orientation as quaternion and time stamps:

Table 14: TPD05

	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
TPD05	LSB x	MSB x	LSB y	MSB y	LSB z	MSB z	LSB time stamp	MSB time stamp

TPD06 contains the acceleration raw data and time stamps:

Table 15: TPD06

	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
TPD06	LSB x-axis	MSB x-axis	LSB y-axis	MSB y-axis	LSB z-axis	MSB z-axis	LSB time stamp	MSB time stamp

TPD07 contains the rotation rate raw data and time stamps:

Table 16: TPD07

	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
TPD07	LSB x-axis	MSB x-axis	LSB y-axis	MSB y-axis	LSB z-axis	MSB z-axis	LSB time stamp	MSB time stamp

8.1.3 Transmission types

The various transmission types of the TPD01 can be set in indexes 0x1800 (TPD01) to 0x1806 (TPD07). These indexes are each comprised of the following subindexes:

Subindex	Name	Permissible values	Description
1	COB-ID	-	The COB ID is automatically adjusted to the note ID and should not be amended by the user.
2	Transmission type	0xFE, 0xFF	The transmission type is set here. The MLS only supports event-driven transmission; the permissible values are 0xFE and 0xFF
3	Not used	-	Not used
4	Compatibility entry	-	Not used
5	Event timer	0 ... 65535	The event timer sets the time between two transmissions from the TPD0 in [ms]. A value of 0 deactivates the transmission of the process data.



NOTE

A small **Event timer** value increases the bus load. A high value reduces the bus load, but correspondingly reduces the track sampling rate, resulting in coarser resolution. This can lead to problems, especially at high speeds.

8.2 Operating and status indicators

Operating indicator

The sensor has a capacitive keypad for performing configuration and setting parameters.

Operation is carried out by pressing a series of keys with various time windows:

Press: Touch the keypad from 0.1 to 0.5 s, then release (> 0.1 s).

Hold: Touch the keypad for several seconds without releasing.

Lift finger: The finger does not touch the keypad for several seconds.



NOTE

Operating the keypad requires a little practice because the response times are limited and the required settings are configured with time dependence.

Tip: Memorize the series for the required settings before you configure the sensor.

A full description of the various configuration options can be found in the chapter "[Teach mode](#)", page 41.

Status indicators

The table below describes the individual function displays. The actual behavior of the LEDs during operation represents a combination of these function displays.

There are two LED windows, each of which has two colors:

- LED window 1: yellow and red
- LED window 2: green and blue

LED window 1:

- Yellow LED on when a magnetic tape with sufficient magnetic field strength is detected (at line level 3 or above)
- Yellow LED flashes when a magnetic field with insufficient field strength is detected (at line level 2 or below)
- Yellow LED off if no magnetic tape is detected
- Red LED on if an error is detected

LED window 2:

- Indicates whether the line has been detected on the left or the right. The blue and green LEDs are dimmed using a software PWM (base frequency 1 kHz, refresh rate 50 Hz; [see figure 17](#)).

If the detected line is on the left-hand side, this is indicated via a dimly illuminated green LED; if the detected line is on the right-hand side, this is indicated via a dimly illuminated blue LED. In position = 0, both LEDs are permanently on.

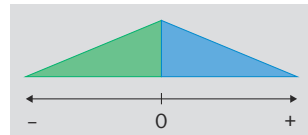


Figure 17: Status indicators

For each color, the full scale ranges from 0 mm → brightly lit | MB/2 | → very dimly lit.

If no magnetic field or multiple lines are detected, this indicator is off. In such cases, the green LED flashes in a special cycle.

The inversion of the position signal (sensor flipped) does not affect the LED behavior.

8.3 Teach-in mode

8.3.1 General information on teach-in mode

In teach-in mode, the current mode is displayed by the LEDs lighting up in a specific way.

8.3.2 Calling up teach-in mode

To activate teach-in mode, press and hold the keypad until the required teach-in mode is reached.

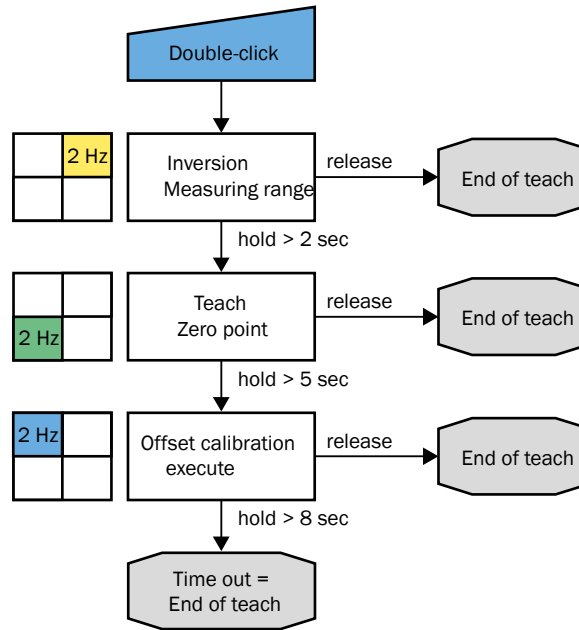


Figure 18: Teach-in sequence

All incorrect teach-in attempts are acknowledged by the sensor with a red flashing LED. If the user attempts to teach in an invalid configuration, the red LED will also flash (e.g., teach zero point with multiple lines).

8.4 Track guidance

8.4.1 Output of line center points

The MLS is capable of detecting up to three line center points (LCPs) and their width (LCP_width), and providing the related data as output via the CANopen interface. The position of each line center point is output to a resolution of 1 mm. The geometric center of the sensor's longitudinal axis is the zero point.

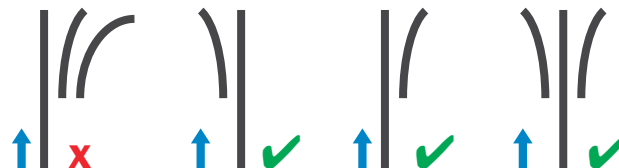
Should the installation space of the AGV not allow for center installation of the sensor, the zero point of the sensor can be moved in the service data (see ["CANopen service data object directory"](#), page 28).



By default, the negative measuring range is towards the cable outlet, and the positive measuring range is on the opposite side.

The alignment of the sensor (positive/negative measuring range) can be changed via the service data.

If only one line center point is found, it is output as **LCP2**. If a further line center point is found, it is output as **LCP1** or **LCP3**, depending on its direction. If three LCPs are found, then each LCP is output.



To make it easier for the control system to evaluate this process byte, the combination of tracks detected is output in an additional data item #LCP. The LCPs are binary-weighted:

Table 17: Line center points

LCP1 detected	LCP2 detected	LCP3 detected	Output value	Comments
No	No	No	0	Special case: No line detected
No	Yes	No	2	Only one line detected
Yes	Yes	No	3	Single diverter detected
No	Yes	Yes	6	Single diverter detected
Yes	Yes	Yes	7	Double diverter detected

The principle of $LCP1 < LCP2 < LCP3$ always applies to the LCPs.

8.4.2 Inversion of the relative position

This function enables the user to invert the convention that the negative range is at the cable outlet. This flips the position by 180° to facilitate sensor installation.

The selected setting affects the output of the inertial measuring unit (IMU), see ["Inertial measuring unit"](#), page 53.

The inversion can be done via CANopen or the teach process on the device.

The inversion of the position signal does not affect the LED behavior.

8.4.3 Accuracy of line detection

In order to guarantee the specified repeatability of +/- 1 mm even at large working distances, a configured average filter ([section 8.4.5.3](#)) is installed in the MLS. In general, the following applies: The greater the working distance, the higher the filter value should be. At the same time, a higher filter value also results in a higher delay time in the signal output. Die t_{90} -time of the step response for various filters settings is listed in the following table:

Table 18: Filter settings

Filter value	t_{90} -time
0	0
1	35 ms
2	57 ms
3	80 ms
4	100 ms

The correlation between track distance, magnetic field strength and filter setting are listed in the following table. The magnetic field strength is output in three parameters:

- **Field Level:** Maximum measured absolute field strength (UINT16, measuring range 25 mT, resolution 0.00076 mT)
- **Line Level:** Field strength for each detected track (INT8, measuring range 6.25 mT, resolution 0.049 mT)
- **Track Level:** Evaluation of the magnetic field strength as 3-bit value in TPD01. For the assignment of Field Level to Track Level see [table 19](#)

Table 19: Assignment table

FieldLevel	LineLevel	Track Level	Magnetic field strength [mT]	Distance to track tape ¹	Recommended filtering
4500	71	7	3.43	10 mm	0

FieldLevel	LineLevel	Track Level	Magnetic field strength [mT]	Distance to track tape ¹	Recommended filtering
2250	35	7	1.72	20 mm	0
2000	31	7	1.53	22 mm	0
1750	28	6	1.34	25 mm	0
1500	24	5	1.14	28 mm	0
1250	20	4	0.95	31 mm	0
1000	16	3	0.76	37 mm	1
750	12	2	0.57	44 mm	1
500	8	1	0.38	55 mm	1
450	7	0	0.34	60 mm	2
350	5	0	0.27	70 mm	4

¹ Measured with SICK track tape item 5337613

At an increased working distance (< **Track Level 3**), an offset calibration should be run. Permanent interfering factors by the AGV can be suppressed in this way.



NOTE

Do not run the offset calibration in the presence of the magnetic tape.

8.4.4 Bitmap



NOTE

This function is available from firmware version 5.

The sensor offers the option of digitally outputting the current magnetic field. The measured value of each sensor element is compared to a threshold when doing so. The following table provides an overview of the number of installed sensor elements of the length variants:

MLS variant	Number of sensor elements
MLSE-0200	36
MLSE-0300	54
MLSE-0400	66
MLSE-0500	84
MLSE-0600	102

If the measured value is larger than this threshold, a 1 is assigned to the element, otherwise a 0 is assigned. This bitmap is output via TPD03 (parameter in index 0x1802) or SDO index 0x202E, subindexes 1-4. The following settings can be made:

- **Bitmap_threshold** (index 0x202F, subindex 1): The threshold (default: 500) with which the measured value of the respective sensor element is compared can be set here. This is a UINT16 value.
- **Bitmap mode** (index 0x202F, subindex 2): The following settings are available:

Value	Meaning
0	Bitmap detection is deactivated.
1	Both polarities are detected as valid. A 1 is assigned to the element when the value is larger than the Bitmap_threshold or smaller than the Bitmap_threshold .
2	Only south polarity is detected as valid. A 1 is assigned to the element when the value is larger than the Bitmap_threshold .

Value	Meaning
3	Only north polarity is detected as valid. A 1 is assigned to the element when the value is smaller than the Bitmap_threshold.

8.4.5 Track optimization



NOTE

This function is available from firmware version 5.

8.4.5.1 Offset calibration

In production, the MLS is calibrated freely in the air (not mounted) at the factory.

In this case, the raw data is near the zero line (sensor orientation: label on the upper side).

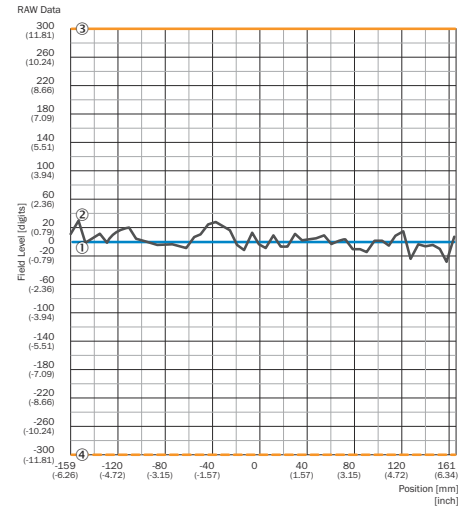


Figure 19: MLS not mounted

If the MLS is mounted in the application (in the AGC/AGV), installation can influence the sensor:

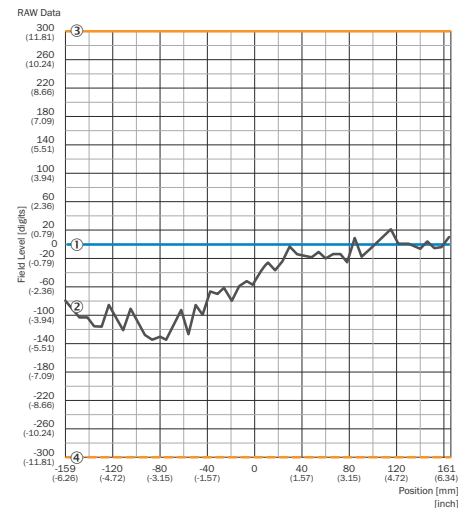


Figure 20: MLS mounted

These influences can be permanently compensated for with an offset calibration. To do so, write value 1 in index 0x202B. The raw data then looks as follows:

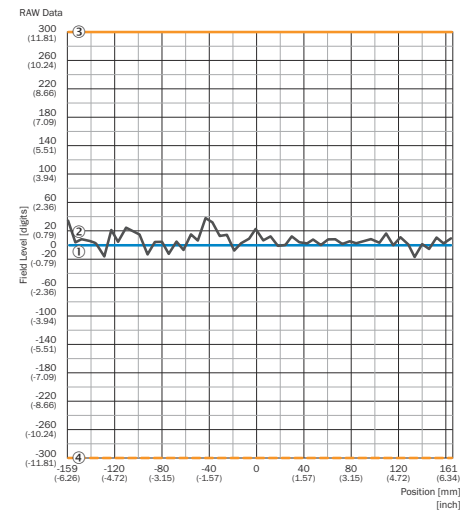


Figure 21: Raw data after offset calibration

- ① Detection Level positive
- ② Raw data
- ③ Min Level positive
- ④ Min Level negative
- ⑤ Detection Level negative

8.4.5.2 Detection Level

If the MLS is placed over the magnetic track, it detects positive and negative signal peaks.

The set Detection Level applies both for the detection of positive and negative signal peaks.

- Positive peaks are detected above the level of the blue solid line (1).
- Negative peaks are detected below the level of the blue dashed line (5).

If the magnetic field has a main maximum (largest absolute value is positive), it should be ensured that the negative Detection Level (5) is far enough UNDER the negative overshoots. In contrast, with a main minimum, the positive Detection Level (1) must be high enough OVER the positive overshoots. This prevents unwanted error detection of tracks or markers.

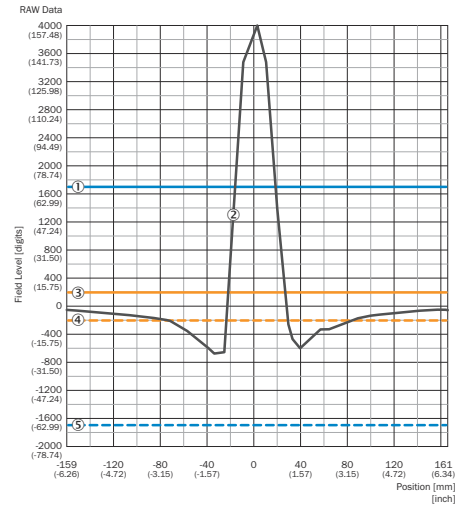


Figure 22: Magnetic field with main maximum

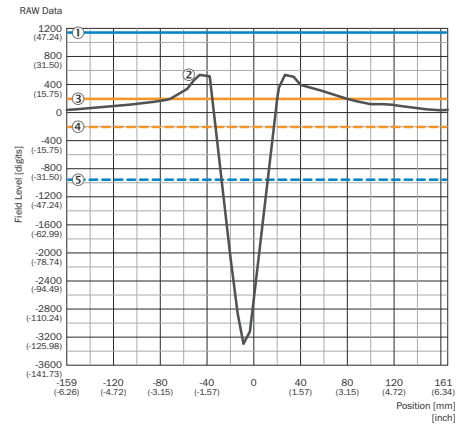


Figure 23: Magnetic field with main minimum

The Detection Level of the MLS can be adjusted using the **First Level** (index 0x202D.1) and **Last Level** (0x202D.2) parameters. The currently implemented algorithm of the MLS works with only one level, which is why: **First_Level = Last_Level**. The value is given in percent of the peak level. The actual Detection Level is calculated as follows:

Detection Level = (Field Level – Min Level)*First Level + Min Level

In addition, the **Min Level** (3)/(4), which defines the threshold for detection of a track, can be moved (index 0x2025). The parameters can be best optimized when two tracks are very close to one another (e.g. with a diverter). The negative overshoots can lead to problems (figure 24) which can be prevented by a higher (absolute) **MinLevel** (figure 25).

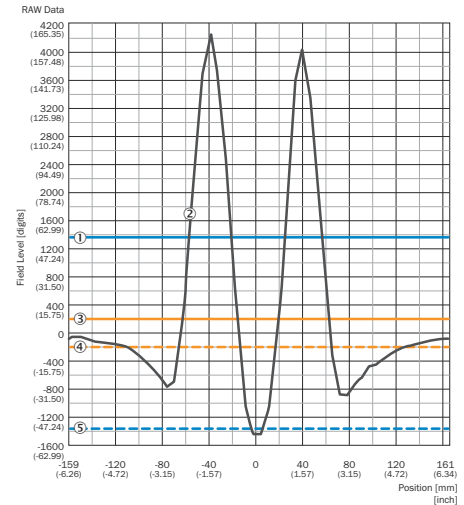


Figure 24: Min Level

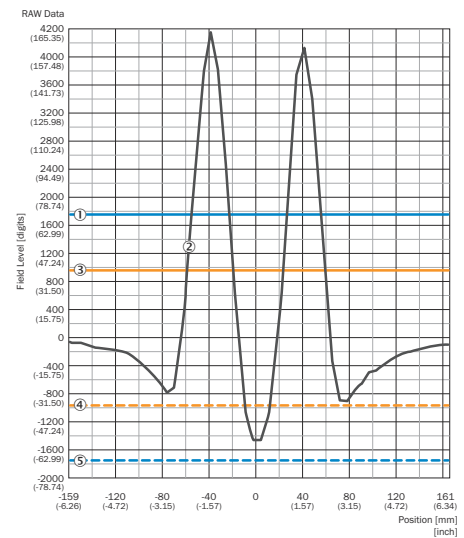


Figure 25: Moved Min Level

- ① Detection Level positive
- ② Raw data
- ③ Min Level positive
- ④ Min Level negative
- ⑤ Detection Level negative

8.4.5.3 Average filter for track output values

Another optimization of the track detection is possible thanks to the **averaging magnetic** function. This function encompasses the filtering of the LCP (Line Center Point), Line Width and Line Level track parameters. The filter value is calculated as follows:

$$\text{Value_filtered}(n) = (\text{value_filtered}(n-1) * \text{<averaging magnetic>} + \text{value}(n)) / (\text{<averaging magnetic>} + 1)$$

Value_filtered(n): Currently filtered value

Value_filtered(n-1): Old filtered value

Value(n): Current (unfiltered) measured value

The averaging_magnetic filter value = 0 deactivates this function.

The filter value is configured via index 0x202D, subindex 3:

Table 20: Index 0x202D, subindex 3

Index	Sub	R/W	Object name	Default value
0x202D	3	R/W	Averaging magnetic	1

In general, the following applies: The greater the working distance, the higher the filter value should be. At the same time, a higher filter value also results in a higher delay time in the signal output (see table 18, page 43).

8.4.5.4 Track polarity setting

The track polarity of the track tape used can be fixed to increase the reliability of the sensor in challenging applications. The following setting options exist (index 0x202D, subindex 5):

Value	Meaning
0	The track polarity is automatically detected, both track polarities are valid. Both south tracks (with north markers) and north tracks (with south markers) are detected (standard setting).
1	Only tracks with north polarity are detected. Markers must have south polarity.
2	Only tracks with south polarity are detected. Markers must have north polarity.

8.4.6 General notes on track guidance

The following descriptions are for track guidance purposes only. For newly installed magnetic tape, the notes in section 5.3 must also be taken into account.

Diverters

Diverters can be driven over both flush and non-flush. Here, the different effects on reliability, driving behavior and the recommended sensor setting should be taken into account.

Table 21: Diverters

Type of diverter	Non-flush	Flush
		
Reliability	Reliable	More susceptible to misinterpretation
Driving behavior	Rough track change: Track is detected abruptly	Soft transition
Recommended sensor setting	Improved diverter detection from (Index 0x2006, Subindex 1))	Improved diverter detection at (Index 0x2006, Subindex 1))

More information on improved diverter detection:

Including the measured line width makes it possible to detect diverters early on and prevent jumps in line center point output.

**NOTE**

When switching the sensor on, it should not be located in the area of a diverter and should be at a 90° angle to the track if possible. Switching on can occur without a track.

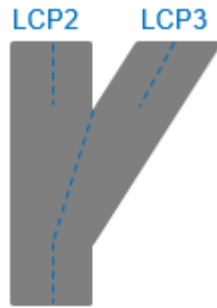


Figure 26: Behavior with standard setting

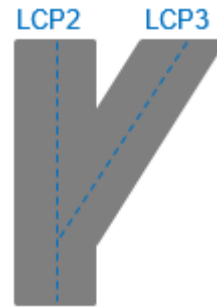


Figure 27: Behavior with improved track detection on diverters

The improved track detection on diverters can be deactivated via Index 0x2006, Subindex 1, value 0. The setting options of the TPDO 1 can be viewed in [section 8.1.2](#). The behavior at diverters then corresponds to firmware version 2 and is recommended with non-flush diverters.

Intersections

Ideally, intersections are identified as follows: #LCP = 7; LCP1 = LCP2 = LCP3. Since this case rarely occurs due to very small tolerances and also cannot be guaranteed, navigation should be done via markers and not via LCP=7.

Curves

When navigating on curves, paying attention to the following parameters is advised:

- Max. AGV speed
- MLS event timer
- MLS measuring range
- Turning circle = min. curve radius of the AGV

The spatial resolution at maximum traversing speed can be calculated from the maximum speed of the AGV and the set event timer of the MLS:

$$\text{Spatial resolution [mm]} = \text{MaxSpeedAGV [m/s]} * \text{EventTimer [ms]}$$

The spatial resolution can be used to calculate the positional deflection of the first sample in the curve at a defined curve radius. This first sample must in any case be within the measuring range of the MLS. If this first sample is no longer in the measuring range, the curve is overrun and the track is lost.

In addition, the minimum curve radius must be larger than the turning circle of the AGV. In summary, the following statements describe the behavior when navigating on curves:

- The higher the traversing speed, the larger the curve radius
- The smaller the event timer, the smaller the curve radius
- The larger the measuring range, the smaller the curve radius

Speed

The spatial resolution of the position output in the traverse direction can be calculated. This should be considered e.g. for curves, diverters as well as markers. If a diverter has too large an angle or if a marker is too short, these will be “overlooked” due to the insufficient spatial resolution, since the detection situation lies between two successive measured values.

Angle (for diverters)

- The greater the speed, the smaller the angle should be for diverters, whereby the diverter must always remain within the MLS measuring range, depending on the spatial resolution.
- The smaller the speed, the greater the angle can be for diverters, since the magnetic field separates earlier and detection is therefore better.

If improved diverter detection is activated, the behavior can also be optimized via the filter strength (Index 202D.6):

- The greater the speed, the smaller the selected value should be.
- The larger the angle, the smaller the selected value should be.

Interference field suppression

- The interference field suppression can suppress and thus eliminate large-scale interference fields.
- The slope of the magnetic track to be detected must be large enough to distinguish it from the large-scale interference fields with low slope in the magnetic field.
- The slope (difference from one magnetic sensor in the MLS to the next) can be set via Index 202D.2. The slope should be set so that the track is still reliably detected.
- When setting the slope, the following applies: The higher this value, the better the interference field suppression, but only magnetic tracks with high magnetic field strength are detected. The lower this value, the lower the interference field suppression, but tracks with low magnetic field strength are also detected.

Procedure in case of errors during track guidance:

If track guidance problems occur and the sensor does not provide the desired values, then the following steps can be followed to analyze and correct the problem in more detail:

1. Reading out the raw data
The raw data can be read out via the configurator available on the website. These are often helpful for situation-specific problems. Interference fields, misinterpreted values and faulty spots can be identified in this way.
2. Adjusting the **field level** and **detection level**
The desired sensitivity of the sensor can be set by adjusting the **field level** and **detection level**. To do this, view the previously read raw data and the desired output values of the sensor. Make the required settings.
Example: The sensor should detect two tracks with magnetic fields of different strengths simultaneously. Here, better detection can be achieved by lowering the **detection level**.
3. Switching on interference field compensation
If interference fields cause the sensor to supply incorrect information, interference field compensation can be activated via TPDO 1 (further information at [section 8.1.2](#)).

8.5 Marker detection

The MLS is capable of detecting magnetic codes using so-called **markers**. Markers are small strips of magnetic tape placed adjacent to the actual track.

Various types of marker can be detected:

SICK marker detection

SICK marker detection uses one or two markers adjacent to the line, but only 1 marker may be placed on each side.

The system supports both polarities: Track tape with the north pole facing up, marker with the south pole facing up; track tape with the south pole facing up, marker with the north pole facing up. The sensor automatically detects the polarity of the (main) track tape.

The value of the code is determined by the distance between the marker and the track. The markers are placed on a 30 mm raster.

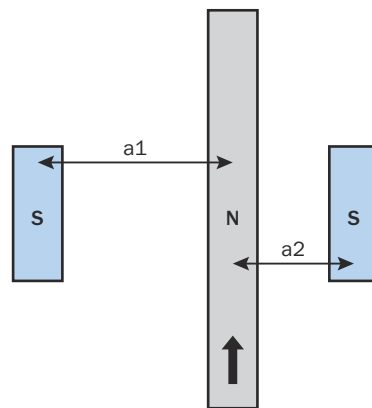
If the main track is 50 mm and wider, directly adjacent markers cannot be used. The distance of 30 mm to the track center can then no longer be realized without overlapping.

A marker length of 100 mm is recommended.



NOTE

For optimum functioning of automated polarity detection, we recommend that the sensor is not switched on over a marker.



The following principle applies:

If $a1 > a2$: Marker detected in positive travel direction

If $a1 < a2$: Marker detected in negative travel direction



NOTICE

The inversion of the position signal (**sensor flipped**) affects marker detection! The position signals after inversion are used to detect the markers!

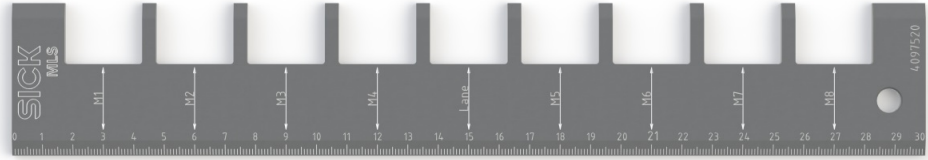
There are a number of different modes:

Standard mode (7 different codes possible): The values of the markers are added:

Marker 1		Marker 2		Total	Code
A1 [mm]	Template ¹	A2 [mm]	-Template ¹		
30	M4	-	-	30	1
60	M3	-	-	60	2
60	M3	30	M5	90	3
90	M2	-	-	90	3
90	M2	30	M5	120	4
120	M1	-	-	120	4
90	M2	60	M6	150	5
120	M1	30	M5	150	5
120	M1	60	M6	180	6

Marker 1		Marker 2			
A1 [mm]	Template ¹	A2 [mm]	-Template ¹	Total	Code
120	M1	90	M7	210	7

¹ Marker template (SICK accessories, part number 4097520)



Extended mode (10 different codes possible):

Marker 1		Marker 2		
A1 [mm]	template ¹	A2 [mm]	template ¹	Code
30	M4	-	-	1
60	M3	-	-	2
60	M3	30	M5	3
90	M2	-	-	4
90	M2	30	M5	5
90	M2	60	M6	6
120	M1	-	-	7
120	M1	30	M5	8
120	M1	60	M6	9
120	M1	90	M7	10

¹ Marker template (SICK accessories, part number 4097520)

FailSafe mode (applies to SICK marker detection only):

FailSafe mode is intended to prevent codes from being read incorrectly if the AGC is not aligned completely parallel to the track and the markers. In such situations, one of the markers may be detected sooner than the marker on the other side. This results in the marker code being misinterpreted. In **FailSafe** mode, the sensor outputs the detected code after it has completely passed over a frame. If **FailSafe mode** is deactivated, the sensor constantly outputs the currently detected marker code. This can cause a code to change or be output incorrectly at a certain time.

8.6 Inertial measuring unit



NOTE

This function is available from firmware version 5.

The line guidance sensor contains an integrated inertial measuring unit (IMU) which enables determination of the current orientation of the sensor. The orientation is output as sensor torsion compared to a reference system which is defined in "[Coordinate system](#)", [page 54](#). The torsion can be represented either by Euler angles or a quaternion.

The IMU is located at the following position (x | y | z): 89.1 | 2.5 | -9.4 [mm]. The zero point shown in [figure 28](#) is located in the center of the edge of the status LEDs.

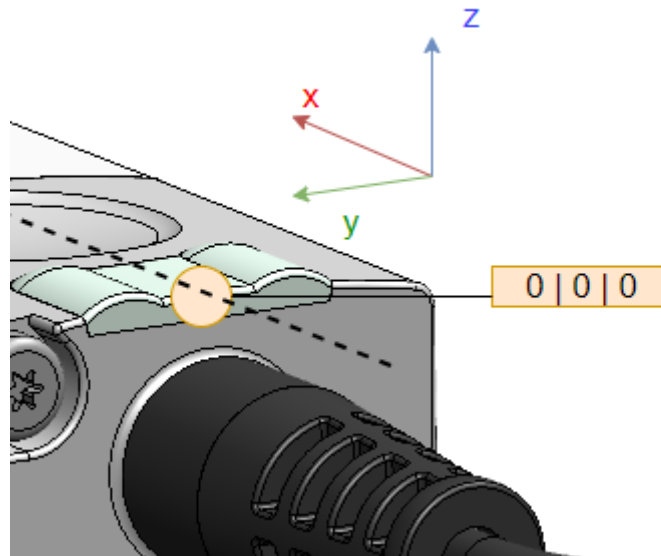


Figure 28: XYZ reference for the IMU position

**NOTE**

The IMU is activated by default and can be deactivated using Index 0x2006, Subindex 2.

Value	Meaning
0	IMU and temperature sensor inactive
1	IMU and temperature sensor active

**NOTE**

The determined rotation of the sensor around the gravity axis (yaw angle) is not stable in the long term due to the principle.

Tip: Only use the yaw angle for relative measurements over limited periods of time. It can be reset to zero before a measurement, see ["Resetting the yaw angle"](#), page 56.

**NOTE**

During its initialization, which runs in 640 ms after start-up of the sensor, the inertial measuring unit does not output valid data (all output values are zero at this time).

8.6.1 Coordinate system

During orientation measurement, the sensor torsion is determined compared to a reference system. The coordinate system used depends on the value of the **sensor flipped** (Index 0x2027) parameter. For the **sensor flipped** = 0 setting, the coordinate system is illustrated as in the following figure:



Figure 29: Standard axis definition

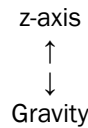
For the **sensor flipped = 1** setting, on the other hand, the following coordinate system is used:



Figure 30: Flipped axis definition

Reference system:

In the reference system, the z-axis is antiparallel to the gravity vector.



The zero point for torsion of the reference coordinate system around the z-axis results from the orientation of the sensor at the power-up delay time (or at the time at which the respective alignment was reset to zero, [see "Resetting the yaw angle", page 56](#)).

8.6.2 Time stamp

A sample timestamp in Index 0x2035 is provided for the measured values. This specifies the measuring time in milliseconds. From the output UINT16 value, the sample time can be calculated as

$$\text{TIME} = 1/1,000 \text{ s} \cdot \text{TIMESTAMP_UINT16}$$

The sample time stamp runs to zero every 65.536 s.

8.6.3 Output of orientation in Euler angle representation

The current orientation in Euler angle representation can be accessed in Index 0x2030, Subindex 1 – 3, whereby the three subindexes contain the Roll, Pitch and Yaw components. Alternatively, the angles can be output in TPDO4. In addition to the three angles, the TPDO contains the time stamp sample.

The angles specify the rotation of the sensor compared to the reference position according to DIN 9300 / DIN ISO 8855 in Z/Y'/X" sequence (intrinsic) or X/Y/Z (extrinsic).

The Euler angles are output in radians. The output INT16 value can be converted to the angle in radians as follows:

$$\text{ANGLE} = 1/10,000 \text{ rad} \cdot \text{VALUE_INT16}$$

8.6.4 Output of orientation in quaternion display

The current orientation in quaternion representation can be accessed in Index 0x2031, Subindex 1–4, whereby the four subindexes contain the real part and the three imaginary parts of the quaternion. The output quaternion is standardized to value 1 and the real part of the quaternion is always positive.

Alternatively, the quaternion can be output in TPDO5. This TPDO contains the imaginary parts of the quaternion and the sample time stamp. Due to the standardization, the real part of the quaternion (not contained in the TPDO) can be calculated as the imaginary parts with

$$w = \sqrt{1 - (x^2 + y^2 + z^2)}$$

x, y and z and the real part with w.

The components of the quaternions do not have units. The value of the components is output as the INT16 value, the actual value can be calculated as

$$\text{COMPONENT} = 1/10000 \cdot \text{VALUE_INT16}$$

8.6.5 Resetting the yaw angle

The output yaw angle is not stable in the long term due to the principle of its measurement. That is why it can be reset to zero by writing the value 1 into the SDO **reset yaw** (Index 0x2032, Subindex 1).

8.6.6 Low pass filter for orientation

The output values for the orientation can be filtered with an adjustable IIR low pass of the first order, whereby the filter affects the Euler angle and the quaternion representations in the same manner.

The filter is configured via filter parameter K in Index 0x202D, Subindex 4, which can be set to whole values between 0 and 100. The resulting -3 dB limit frequency for the output signals can be calculated as

$$f_{-3dB} = 31,83\text{Hz} \cdot \cos^{-1} \left(1 - \frac{1}{2K \cdot (K + 1)} \right)$$

Higher values for K therefore result in stronger low pass filtering of the signal. For a value of 0 for K, the low pass filter is deactivated (default setting).

8.6.7 Output of raw data for acceleration and yaw rate

For acceleration and yaw rate, the raw data can be called up in index 0x2033, subindex 1–3 (acceleration) and index 0x2034, subindex 1–3 (yaw rate). The raw data can also be output as TPDOs, the acceleration in TPDO6 and the yaw rate in TPDO7. In addition to the respective raw data, both TPDOs contain the respective sample time stamp.

The coordinate system for the raw data is the system shown in section ["Coordinate system", page 54](#) for **Sensor Flipped** = 1. The selected **Sensor Flipped** setting has no influence on the raw data (these are real, uncorrected raw values).

The raw data has the following parameters:

- Sampling rate: 200 Hz
- Measuring range:
Acceleration: $\pm 16 \text{ g}$
Yaw rate: $\pm 2,000^\circ/\text{s}$

The raw data is output as INT16 values.

The measured acceleration in g can be calculated from this as

$$\text{ACCELERATION} = 2^{-11} \text{ g} \cdot \text{VALUE_INT16}$$

or the acceleration in m/s^2 as

$$\text{ACCELERATION} = 4.79 \times 10^{-3} \text{ m/s}^2 \cdot \text{VALUE_INT16}$$

The measured yaw rate in $^\circ/\text{s}$ can be calculated from the INT16 values as

$$\text{YAW RATE} = 125 \times 2^{-11}^\circ/\text{s} \cdot \text{VALUE_INT16}$$

or the yaw rate in rad/s as

$$\text{YAW RATE} = 1.065 \times 10^{-3} \text{ rad/s} \cdot \text{VALUE_INT16}$$

8.7 Temperature sensor



NOTE

This function is available from firmware version 5.

The temperature currently measured in the sensor can be accessed in Index 0x2070, Subindex 1. The temperature values have data type INT8 and represent the temperature in °C.



NOTE

The temperature sensor is activated/deactivated by activating/deactivating the IMU in Index 0x2006, Subindex 2.

8.8 Event detection



NOTE

This function is available from firmware version 5.

The line guidance sensor can detect two types of events:

- **Magnet field:** This event is triggered when the critical magnetic field strength is not reached. When the critical magnetic field strength is not reached, it can no longer be guaranteed that the unfiltered track position meets the data sheet specifications.
- **Temperature:** This event is triggered when the current temperature exceeds the temperature set in Index 0x2070, Subindex 6 or falls below the temperature threshold set in Index 0x2070, Subindex 7 **lower temperature threshold** (4,016).

When the event occurs, the “Event” flag is set in the status byte in TPD01 and in Index 0x2022. The cause of the event can be accessed in Index 0x2080, Subindex 2. The assignment is shown in table [table 22](#).

Table 22: Event source

Value in Event source	Cause of event
0	No event
2	Magnetic field
4	Temperature
6	Magnetic field and temperature

Further details of the cause of the event can be accessed in Index 0x2080, Subindex 4 for a magnetic field event, and Index 0x2080, Subindex 5 for a temperature event. The possible error codes are shown in table [table 23](#) and [table 24](#).

Table 23: Magnetic field error code

Magnetic field error code	Cause of event
0	No event
1	Magnetic field falls below critical magnetic field strength

Table 24: Temperature error code

Temperature error code	Cause of event
0	No event

Temperature error code	Cause of event
1	Current temperature is below the lower threshold (Index 0x2070, Subindex 7)
2	Current temperature exceeds the upper threshold (Index 0x2070, Subindex 6)

The event flag can be reset by writing to Index 0x2080, Subindex 1.

9 Maintenance

9.1 Maintenance

The sensor is maintenance-free.

To ensure it continues operating without problems, however, the screw connection between the sensor and the slot, and for the electrical connection, should be checked regularly. The interval at which they are checked should be adapted in line with the conditions of the application, but should be no more than 6 months.

9.2 Repairs

Repairs on the sensor may only be carried out by the manufacturer. Any interruption or modification of the sensor will invalidate the manufacturer warranty.

10 Decommissioning

10.1 Decommissioning

Removing the sensor

1. Switch off the supply voltage to the sensor.
2. Detach all connecting cables from the sensor.
3. If the sensor is being replaced, mark its position and alignment on the bracket or surroundings.
4. Remove the sensor from the slot.

Disposing of the sensor

Any sensor which can no longer be used must be disposed of in an environmentally friendly manner in accordance with the applicable country-specific waste disposal regulations. The sensor is electronic waste and must under no circumstances be disposed of with general waste.

11 Troubleshooting

Possible faults and corrective actions are described in the table below for troubleshooting. In the case of faults that cannot be rectified using the description below, please contact the manufacturer. See the back page for relevant contact details.

Table 25: Troubleshooting

LED indicator/fault pattern	Cause	Measure
No LED illuminated/sensor not communicating via CAN-open	Problem with sensor voltage supply	Check and restore voltage supply
Green LED flashing/sensor does not output track data even though track tape is positioned at the correct distance	Problem with sensor voltage supply	Reduce threshold for detection of track tape (Index 0x2025)
Yellow LED lights up/sensor does not output track data	Event timer TPD02 is set to '0'	Set event timer (index 0x1800 subindex 3) to a value greater than 10
Sensor does not send a TPD02-7	TPD02-7 not activated	Activate TPD02-7 via COB ID

12 Technical data

Performance

Table 26: Performance

Parameter	Conditions	Icon	Unit	min.	Typ.	max.
Measuring range		MR300	mm		300	
Repeatability ¹			mm			1
Resolution			mm			1
Working distance ²			mm	10		70 ⁴
Output refresh rate ³			Hz			100
LED display				Yellow / red, blue / green		
Reverse polarity protection				Yes		
Short-circuit protection				Yes		

¹ Depending on magnetic tape used and working distance

² Depending on magnetic tape used

³ Depending on sensor length:

200: 100 Hz

300: 100 Hz

400: 80 Hz

500: 70 Hz

600: 60 Hz

⁴ With magnetic averaging value "4", see ["Track optimization"](#), page 45

Interfaces

Table 27: Interfaces

Parameter	Specification
Interface	CANopen
Device profile	No device profile is supported
Address setting	0 ... 127, default: 10
Data transmission rate	10 kBaud ... 1,000 kBaud, default: 125 kBaud
PDO data	Track position and width Line level Marker Status Orientation Euler angle / quaternion IMU raw data
Configuration data	Minimum permissible magnetic field strength, offset zero point, inversion of measuring range, marker detection, activation of teach-in button
Diagnostic data	Current magnetic field strength
Status information	No LED allocated to CANopen bus
Bus termination	Internal 120 Ohm terminator (optional)

Mechanics/Electronics

Table 28: Mechanics/Electronics

Parameter	Specification
Electrical connection	PUR cable 0.3 m; M8 4-pin or PUR cable 0.3 m; M12, 4-pin or PUR cable 2 m; cable connection (open strand end)

Parameter	Specification
Supply voltage	9 ... 30 V DC, reverse polarity protected
Residual ripple	</= 10%
Power consumption	</= 600 mW
Housing	Aluminum, PA
Housing color	Black, end caps: Black
EMC	EN 61000-6-2 EN 61000-6-4
Protection class	3
Enclosure rating	IP 65, IP 67, IP 68 (in accordance with EN 60529)
Mounting	Mounted using accessories in sensor T-slot

	Total length	Measuring range
MLSE-0200	217	200
MLSE-0300	325	300
MLSE-0400	397	400
MLSE-0500	505	500
MLSE-0600	613	600

Ambient data

Table 29: Ambient data

Parameter	Specification
Permissible impact load	30 g / 11 ms
Permissible impact load	10 ... 55 Hz / 1 mm
Permissible ambient temperature	-20 °C ... 70 °C

13 Annex

13.1 Conformities and certificates

You can obtain declarations of conformity, certificates and the current documentation for the product at www.sick.com. To do so, enter the product part number in the search field (part number: see the entry in the “P/N” or “Ident. no.” field on the type label).

13.2 Modbus RTU Spezifikationen

For detailed information about Modbus see:

<https://modbus.org/>

We recommend for test purposes or quick parameterization:

<https://sourceforge.net/projects/qmodmaster/>

13.3 RTU Transmission Mode

In RTU mode, messages start with a silent interval of at least 3.5 character times. The most easy way to implement this is as a multiple of character times at the baud rate that is being used on the network (shown as T1-T2-T3-T4 in the figure below). The first field that is transmitted is the device address.

The allowable characters transmitted for all fields are hexadecimal 0-9, A-F. Networked devices monitor the network bus continuously, including during the ‘silent’ intervals. After receiving the first field (the address field), the encoder decodes it to check whether it is the device being addressed.

Following the last transmitted character, a similar interval of at least 3,5 character times marks the end of the message. A new message can start after this interval. The entire message frame must be transmitted as a continuous stream. If a silent interval of more than 1.5 character times occurs before completion of the frame, the receiving device flushes the incomplete message and assumes that the next byte will be the address field of a new message. Similarly, if a new message begins earlier than 3.5 character times following a previous message, the receiving device will consider it a continuation of the previous message. This will cause an error, as the value in the final CRC field will not be valid for the combined messages.

A typical message frame is shown below:

Start	Address	Function	Data	CRC Check	End
T1-T2-T3-T4	8 Bits	8 Bits	n x 8 Bits	16 Bits	T1-T2-T3-T4

13.4 Data Model

The data model of Modbus is based on a series of tables that have distinguishing characteristics. The primary tables are:

Table 30: Primary table

Primary table	Object type	Type of	Comments
Discrete Input	Single bit	Read-Only	This type of data can be provided by an I/O system.
Coils	Single bit	Read-Write	This type of data can be alterable by an application program.
Input Registers ¹	16-bit word	Read-Only	This type of data can be provided by an I/O system.
Holding Registers ¹	16-bit word	Read-Write	This type of data can be alterable by an application program.

¹ MLS supports Input and Holding Registers

13.5 Example Function 04 Read Input Registers

Query

The query message specifies the starting register and quantity of registers to be read. This is an example of a request to read registers 212 – 214 from MLS device:

Table 31: Query

Field Name	Example (Hex)
Device Address	0A
Function	04
Starting Address Hi	00
Starting Address Lo	D4
No. of Points Hi	00
No. of Points Lo	03
Error Check Hi (CRC)	F1
Error Check Lo (CRC)	48

Response

The register data in the response message are packed as two bytes per register, with the binary contents being aligned to the right within each byte. For each register, the first byte contains the high order bits and the second contains the low order bits. The response is returned when the data is completely assembled. This is an example of a response to the previous query:

Table 32: Response

Field Name	Example (Hex)
Device Address	0A
Function	04
Byte Count	06
Data Hi (Register 212)	00
Data Lo (Register 212)	19
Data Hi (Register 213)	00
Data Lo (Register 213)	19
Data Hi (Register 214)	00
Data Lo (Register 214)	19
Error Check Hi (CRC)	1E
Error Check Lo (CRC)	6C

13.6 Example Function 16 (10Hex) Preset Multiple Registers

Query

The query message specifies the register references to be preset. This is an example of a request to preset two registers starting at 4000, in slave device 10 (0x10):

Table 33: Query

Field Name	Example (Hex)
Device Address	0A
Function	10
Starting Address Hi	0F
Starting Address Lo	A0

Field Name	Example (Hex)
No. of Registers Hi	00
No. of Registers Lo	02
Byte Count	04
Data Hi	01
Data Lo	F4
Data Hi	00
Data Lo	05
Error Check Hi (CRC)	1D
Error Check Lo (CRC)	0E

Response

The normal response returns the device address, function code, starting address, and quantity of registers preset. This is an example of a response to the query shown above.

Table 34: Response

Field Name	Example (Hex)
Device Address	0A
Function	10
Starting Address Hi	0F
Starting Address Lo	A0
No. of Registers Hi	00
No. of Registers Lo	02
Error Check Hi (CRC)	43
Error Check Lo (CRC)	85

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